

Goldfinch V1 Documentation

Introduction

What is the Goldfinch Protocol, and why does it matter? Goldfinch's documentation is community-driven.

Bringing real-world investment to DeFi

Goldfinch is a decentralized, globally accessible credit protocol, with a mission to bring the world's credit activity on-chain while expanding access to capital and fostering financial inclusion.

The protocol makes crypto loans without requiring crypto collateral—the missing piece that finally unlocks access to cryptocurrency capital for most people in the world. By incorporating the principle of trust through consensus Goldfinch creates a way for borrowers to show creditworthiness based on the collective assessment of other participants, rather than based on over-collateralizing with crypto assets.

This provides the basis for establishing an immutable, on-chain credit history, a core foundation of any scalable lending model and a primitive that is missing in a meaningful way for many growing markets globally.



The protocol then uses this collective assessment as a signal for allocating capital. The decentralized community of Goldfinch investors make loans to businesses who drive economic growth in their regions, starting with those in emerging markets. Currently, all loans on Goldfinch are fully collateralized with off-chain assets.

By removing the need for crypto collateral and providing a means for automatically distributed yield, the protocol dramatically expands both the potential borrowers who can access crypto and the potential investors who can gain exposure.

Starting with emerging markets

From the beginning, Goldfinch wanted to build for the borrowers who can benefit most and where crypto can have the greatest impact. That meant starting with lending businesses in emerging markets. Demand for crypto's liquid, multinational nature is high in those regions, as businesses stand to gain many benefits from using crypto due to the inefficiencies and barriers of traditional finance limiting the capital that can flow into these regions.

Goldfinch is currently expanding financial access for thousands of individuals around the world via its Borrowers, spanning over 28 countries. Some of Goldfinch's Borrowers include [PayJoy ↗](#) in Mexico, [QuickCheck ↗](#) in Nigeria, [Divibank ↗](#) and [Addem Capital ↗](#) in LatAm, [Greenway ↗](#) through [Almavest ↗](#) in India, and [Cauris ↗](#) in Africa, Asia, and Latin America. You can view more Borrower Pools [here ↗](#).

Read on to the [Goldfinch Overview](#) for a more detailed explanation of the protocol and how it works. You can learn more about Goldfinch's mission and vision in [The Goldfinch Manifesto ↗](#).

For a deeper technical dive into Goldfinch's code, smart contract configuration, security, and other technical features, view the [Goldfinch Developer Documentation site. ↗](#)

Goldfinch Overview

A short overview of Goldfinch's key mechanics

This overview is a condensed and accessible version of the [Goldfinch Whitepaper Version 1.1](#), published in July 2021, which provides the technical and in-depth explanation of the protocol's foundations. You can learn about the below in more detail in the [protocol mechanics section](#) of the documentation.

For a deeper technical dive into Goldfinch's code, smart contract configuration, security, and other technical features, view the [Goldfinch Developer Documentation](#).

Goldfinch is a decentralized credit protocol that allows for crypto borrowing without crypto collateral—with loans instead fully collateralized off-chain.

A core limitation of current decentralized crypto lending protocols is that they require borrowers to overcollateralize their loans with crypto assets, which prevents the vast majority of borrowers in the world from participating.

By incorporating the principle of “trust through consensus,” the Goldfinch protocol creates a way for borrowers to show creditworthiness based on the collective assessment of other participants rather than based on their crypto assets. Currently, all loans on the protocol are fully collateralized with off-chain assets and income.

Participants

The Goldfinch protocol has three key roles: Investors, Borrowers, and Auditors. In addition, Members are Investors that have enhanced their participation to support the protocol's growth. You can learn in more detail about these roles and their incentives (APYs, rewards, etc.) in the [protocol mechanics section](#) of the documentation.

Investors are participants who provide USDC to the protocol to be utilized by Borrowers. There are two ways to become an Investor on Goldfinch: as a Backer or as a Liquidity Provider.

- [Backers](#) optimize for yield and specificity. They assess individual Borrower Pools, decide whether to invest in them directly with first-loss capital, and earn the protocol's highest yields for doing so.
- [Liquidity Providers](#) optimize for diversification and liquidity. They supply second-loss capital to the Senior Pool, which automatically allocates its funds across all Borrower Pools according to the assessment of Backers.

[Members](#) are participants who supply capital and GFI to a Goldfinch Membership Vault to support the network's growth and security, and receive Member Rewards for their enhanced participation.

[Borrowers](#) are participants who seek financing from Goldfinch, and they propose Borrower Pools to be assessed by the network. Borrower Pools are smart contracts that contain the terms a Borrower seeks for their loan, such as the interest rate and repayment schedule.

[Auditors](#) vote to approve Borrowers, a required step before they can propose a Borrower Pool to Backers. Auditors are randomly selected by the protocol to provide a human-level check guarding against fraudulent activity and earn rewards in exchange for conducting this work. Note that as of June 2021 the Auditor role is not yet live on Goldfinch.

How Goldfinch works

Borrowers (currently off-chain lending businesses) propose deal terms for credit lines (Borrower Pools) to the protocol. Goldfinch's community of Investors can then supply capital to these credit lines (Pools), either directly to individual Pools (as Backers) or indirectly by automatically allocating capital across the protocol (Liquidity Providers via the Senior Pool).

These Borrower businesses use their credit lines to draw down stablecoins, specifically USDC, from their Pool. Borrowers then exchange the USDC for fiat currency and deploy it on the ground to end-borrowers in their local markets.

In this way, the protocol provides the utility of crypto—specifically, its global access to capital—while leaving the actual end-borrower loan origination and servicing to the businesses best equipped to handle that in their own communities.

Trust through consensus

In order to determine how to allocate capital from the Senior Pool, the protocol uses a principle of "trust through consensus." This means that while the protocol doesn't trust any individual Backer or Auditor, it does trust the collective actions of many of them.

At a high level: when more Backers supply to a given Borrower Pool, the Senior Pool [increases the ratio with which it adds leverage](#). Because this approach relies on counting individual Backers, the protocol must ensure they are in fact represented by different people. Therefore, all Backers, Borrowers, and Auditors require a "unique entity check" to participate (see below).

Junior and senior tranches

Borrower Pool smart contracts have both a junior and senior tranche, which together account for 100% of a Borrower Pool's capital. Backers supply capital to the Pools' junior tranches, and the Senior Pool supplies capital to the Pools' senior tranches.

When a Borrower makes repayments to one of their Borrower Pools, the Borrower Pool applies the payment first toward any interest and principal owed to the senior tranche at that time, and then toward any interest and principal owed to the junior tranche at that time.

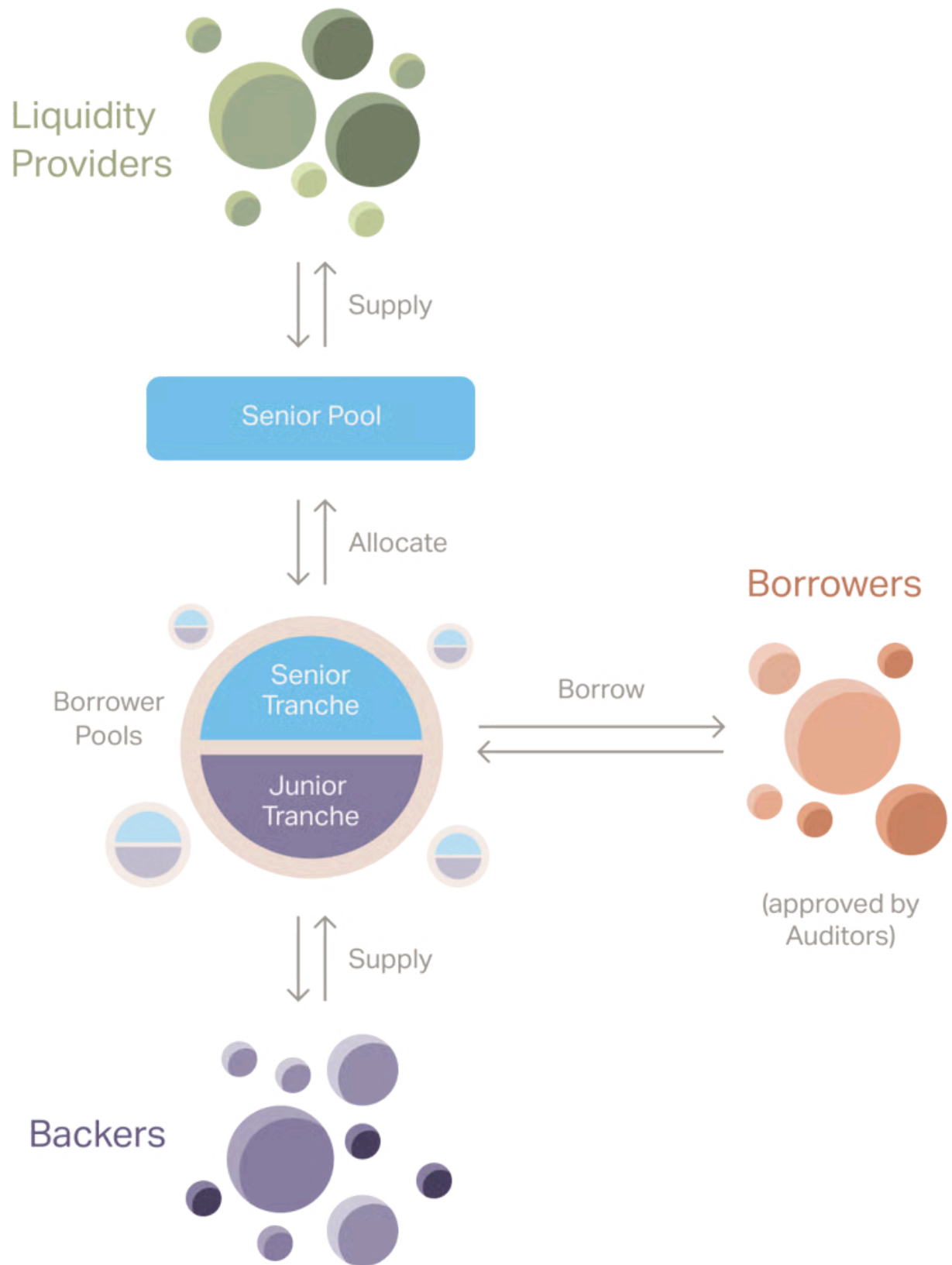
This is to ensure an alignment of incentives. Backers, who actively assess the viability of individual Borrower Pools, take on the protocol's highest risk by providing first-loss capital via junior tranches. This incentivizes Backers to do an adequate job of assessing the viability of the Pools, as they will be the first to miss repayment in the case of a default.

Liquidity Providers, who provide capital to the Senior Pool that is automatically allocated across Borrower Pools according to the actions of Backers, take on less risk by providing second-loss capital via senior tranches. This provides Liquidity Providers with the added security of knowing that they will be the first repaid if a Pool that was evaluated positively by Backers does go into default.

To track the different amounts that investors supply, Backers receive an NFT representing their deposit when they supply capital. The NFT tracks the amount that was supplied and how much of it has been redeemed. This allows the protocol to ensure that no one redeems for more than their proportional share of the total repayments as they come in.

Senior Pool LPs receive FIDU, an ERC-20 token representing their position that grows in USDC exchange value as interest and repayments are made to the Senior Pool.

At any time, a Backer can use their NFT to redeem their specific portion of the available repayments in the Pool. Senior Pool Liquidity Providers can withdraw their position in the Senior Pool at any time by depositing their FIDU to a [Withdrawal Request](#) ↗.



Goldfinch protocol architecture

Tokens

There are two native tokens on Goldfinch: GFI and FIDU. Both follow the ERC20 standard. In addition, the protocol utilizes stablecoins, currently only [USDC ↗](#), for investment and loans.

GFI is Goldfinch's core native token. GFI is used for governance voting, Auditor staking, Auditor vote rewards, community grants, staking on Backers, protocol incentives, and can be deposited into a Member Vault to earn Member Rewards in exchange for helping to secure the protocol's growth.

FIDU is a token that represents a Liquidity Provider's deposit to the Senior Pool. When a Liquidity Provider supplies to the Senior Pool, they receive an equivalent amount of FIDU. FIDU can be redeemed for USDC in the Goldfinch dapp at an exchange rate based on the net asset value of the Senior Pool, minus a 0.5% withdrawal fee. The exchange rate for FIDU increases over time as interest payments are made back to the Senior Pool.

Key Borrower repayment incentives

- Because Borrowers need to publicize their wallet address when proposing pools to Backers, their on-chain history becomes public to future creditors—even those off-chain. As more of global finance moves on-chain, creating a negative on-chain credit history is comparable for future credit opportunities to creating a negative off-chain credit score or record.
- Borrowers will likely want to continue borrowing from Goldfinch. The moment they are late on any payment, Borrowers are unable to borrow further from any Borrower Pool.
- While not explicitly supported by the protocol, Backers may form off-chain legal agreements with Borrowers, including requiring Borrowers to collateralize their on-chain loan with off-chain collateral. Backers may require such an agreement to be in effect, either with them directly or with another Backer, in order to be willing to supply capital. In these cases, the legal agreement and potential Investor recourse are another important incentive for Borrowers. Currently, all loans on Goldfinch are fully collateralized with off-chain assets via this mechanism.

- While occasional defaults are common across finance, Backers will likely stop supplying more capital to any of a Borrower's pools if the Borrower is consistently late on repayments. It is up to Backer to evaluate whether Borrowers are qualified and have a good enough track record to garner Backer investment in any future pools the Borrower proposes.

Governance

Goldfinch is a decentralized, community-driven and community-managed protocol. Governance is managed by the community DAO and has the ability to perform maintenance functions and parameter adjustments via governance votes, including:

- Upgrading contracts
- Changing protocol configurations and parameters
- Selecting Unique Entity Check providers
- Setting rewards and the distribution of GFI
- Pausing protocol activity in the event of an emergency

Unique Entity Check

Since the protocol relies on "trust through consensus," it is critical to avoid [Sybil Attacks](#)  by having confidence that each Borrower, Backer, and Auditor is a unique entity. Therefore, they must each be verified with a "Unique Entity Check" before they can participate.

Governance approves the protocol's Unique Entity Check providers. Currently, this is solved by [Unique Identity \(UID\)](#), the world's first NFT for identity verification.

UID is a non-transferrable NFT representing Know-Your-Customer (KYC), Know-Your-Business (KYB), and/or U.S. investor accreditation verification on-chain. It follows ERC-1155 standards, and is freely usable by any other protocol. No personally identifiable data is stored on-chain. UID is a product developed by Warbler Labs, the core development team supporting Goldfinch's growth.

At any time the Goldfinch community can use the governance process to update or change how the Unique Entity Check is being conducted, such as by introducing oracles that perform off-chain checks to validate that the wallet addresses are unique entities.

Ready to learn more?

This brief overview is only intended to provide a baseline understanding of Goldfinch and how it functions. To learn about the above in more detail, along with additional information on incentives and rewards, fraud resistance, the Leverage Model that determines Senior Pool allocation, and more, read on through the [Protocol Mechanics sections](#).

Protocol Mechanics

Glossary of key terms

Participants

Auditors — Participants who vote to approve Borrowers based on on- and off-chain evaluation, a required step before they can borrow. Auditors receive GFI rewards for securing the protocol with a human eye.

Backers — Investors who supply capital to individual Borrower Pools. Backers evaluate individual deals and lend directly to specific Borrower Pools via the junior tranche (first-loss).

Borrowers — Participants who raise capital from the protocol via Borrower Pools.

Investors — Participants who supply crypto capital to the Goldfinch protocol, either as a Backer or as a Liquidity Provider.

Liquidity Providers — Investors who supply capital to Goldfinch by depositing in the Senior Pool. This Senior Pool capital is automatically allocated across Borrower Pools via the senior tranches (second-loss) according to the Leverage Model.

Members — Participants who supply capital and GFI to a Goldfinch Membership Vault to support the network's growth and security, and receive Member Rewards for their enhanced participation.

Mechanics

Borrower Pool — Investment pools for specific Borrower deals. A Borrower Pool is a smart contract that encodes a set of financing terms for a Borrower, including the interest rate and repayment schedule, proposed by the Borrower, and through which the Borrower can borrow capital and repay it by those terms.

FIDU — A token that represents a Liquidity Provider's deposit to the Senior Pool. When a Liquidity Provider supplies to the Senior Pool, they receive an equivalent amount of FIDU. FIDU can be redeemed for USDC in the Goldfinch dapp at an exchange rate based on the net asset value of the Senior Pool, minus a 0.5% withdrawal fee, a rate which increases over time as interest payments are made back to the Senior Pool. FIDU follows the ERC20 standard.

GFI — Goldfinch's core native token, used for governance votes, Auditor staking, Auditor vote rewards, staking on Backers, and protocol incentives. GFI follows the ERC20 standard.

Governance — Smart contract that is managed by the community DAO and has the ability to update the protocol via decentralized governance votes.

Junior Tranche — The junior (first-loss) capital segment of the Borrower Pool smart contract, funded by individual Backers.

Leverage Model — A formula by which the Senior Pool automatically determines how much capital to allocate to the senior tranche of each Borrower Pool.

Membership Vault — A vault for locking capital to help secure the growth and security of the Goldfinch network as a Member, earning a share of the protocol's fee revenue for doing so in the form of Member Rewards.

Senior Pool — An investment pool for automatically diversified yields. The Senior Pool is a smart contract that accepts capital from Liquidity Providers and automatically allocates capital to the senior tranches of Borrower Pools, according to Goldfinch's Leverage Model.

Senior Tranche — The senior (second-loss) capital segment of the Borrower Pool smart contract, funded by the Senior Pool according to the Leverage Model.

Incentives

Backer Bonus — The additional \$GFI return earned uniquely by Backers, in exchange for taking on the risk of providing the protocol's junior capital as well as for their work evaluating Borrower Pool deals. The Backer Bonus is generated by the Pool's interest repayments.

Curve LP Rewards — GFI rewards for staking Curve FIDU-USDC LP tokens on the Goldfinch protocol.

Early Backer Airdrop — A GFI reward provided to Backers who contributed to a Pool before the implementation of the community governance proposals for Backer Rewards and Backer Staking Rewards mechanisms.

Investor Rewards — The additional GFI return all Investors on Goldfinch receive in exchange for their participation in the protocol. Investor Rewards are an estimated return as they are not defined by Borrowers' financing terms, but are instead dependent on tokenomics and network dynamics.

Member Rewards — Goldfinch Members receive yield enhancements via Member Rewards, which have been earmarked from the Goldfinch treasury and are distributed pro-rata based on one's Membership Vault position.

USDC APY — The base-level USDC return an Investor receives for participating in Goldfinch, generated from Borrowers' interest payments on their loans. The USDC APY is defined in the Borrowers' financing terms when they establish Borrower Pools. For Backers, this APY is a fixed rate as defined by the Borrower Pool's terms and the Senior Pool leverage. For Liquidity Providers, this rate is estimated as the Senior Pool's USDC returns vary based on the Senior Pool's usage and balance.

Borrowers

- ✓ Borrowers are participants who seek financing from the protocol. They propose terms to Backers to supply capital to their Borrower Pools.

Borrower Pool creation

A Borrower Pool is the smart contract through which Borrowers borrow and repay capital. Once approved by Auditors, any Borrower can create a Borrower Pool and define the terms they want:

- **Interest Rate:** Fixed interest rate APR, e.g. 15%.
- **Limit:** Total capital that can be borrowed, e.g. \$10M.
- **Payment Frequency:** Frequency of interest and principal payments, e.g. every 30 days.
- **Term:** The length of time until the full principal is due, e.g. 365 days.
- **Late Fee:** Additional interest owed when payments are late, e.g. 5%.

Creating a Borrower Pool is like proposing a “term sheet” to Backers. It does not guarantee the terms will be accepted, since Borrowers then need to convince Backers to supply junior tranche (first-loss) capital to the Pool. The amount Borrowers can borrow is based on how much Backers supply, combined with the correlated amount the Senior Pool allocates based on the Leverage Model.

Notably, Borrowers need to set a limit for their Borrower Pools: a self-imposed cap on how much capital they can borrow, also referred to as the Pool limit. While Borrowers might ideally want an infinite limit, Backers want to know that they are only staking first-loss capital toward a total potential amount that the Borrowers can safely deploy. Borrowers therefore have an incentive to set the limit only as high as they can convince Backers that they can safely use and repay.

In order to create a Borrower Pool, the Borrower must also stake an amount of GFI equal to double the cost of an Auditor approval, which is a fixed rate set by the protocol and determined by community governance. This helps guard against spam, signals to Backers that the Borrower is serious, and provides GFI to pay for the first Auditor approval. The first half of the staked GFI is used for the first Auditor approval. The Borrower can redeem their remaining staked GFI when they have fully repaid their borrowed balance.

Borrowing and repaying

Borrowers can borrow capital through the Borrower Pool at any time. The total amount they can borrow is the minimum of:

- A. The calculated limit based on the capital that Backers have supplied and the capital automatically allocated by the Senior Pool according to the Leverage Model.
- B. The combined total capital that Backers have supplied in that Borrower Pool plus the remaining capital in the Senior Pool.
- C. The Borrower Pool's limit.

After borrowing, Borrowers make repayments to the Borrower Pool according to its interest rate and payment period. When they pay more than the interest owed, the remainder is applied to the principal balance.

Junior and senior tranches

Borrower Pool smart contracts have both a junior and senior tranche. Backers supply capital directly to the junior tranche, and the Senior Pool supplies capital to the senior tranche. When a Borrower makes repayments, the Borrower Pool applies the amount first toward any interest and principal owed to the senior tranche at that time, and then toward any interest and principal owed to the junior tranche at that time.

To track the different amounts that different participants supply, both the Backers and the Senior Pool receive an NFT when they supply capital. The NFT tracks the amount that was supplied and how much of it has been redeemed. At any time, a Backer or the Senior Pool can use their NFT to redeem their specific portion of the available repayments in the Pool.

The Borrower Pools use NFTs rather than fungible tokens because it allows the protocol to ensure that no one redeems more than their proportional share of the total repayments as they come in.

For example, let's say two Backers have each supplied \$500 for a total of \$1,000 borrowed, and that so far the Borrower has made repayments totaling \$300. In this scenario, the NFTs ensure each Backer can only redeem up to \$150, which is their portion of the repayments so far, rather than each one racing to redeem the full \$300 for themselves.

Origination fee

There may be certain participants who work with Borrowers to establish terms and bring them to the protocol. To compensate them for these efforts, Borrower Pools support an origination fee that is paid to the Pool's originator.

The origination fee is defined as a percentage of the interest. For example, for a \$1M Borrower Pool with 15% interest paid monthly and a 10% origination fee, the Borrower would pay monthly interest of \$12.5K and the originator would receive a monthly fee of \$1.25K. To align incentives with Investors, the originator fee is treated as the most junior tranche, so every payment first goes toward what is owed to the Senior Pool (senior tranche) and Backers (junior tranche) before it goes toward the originator fee.

Borrower repayment incentives

✔ A key question is what incentives Borrowers have to pay back what they borrow

The first incentive is the establishment of a positive on-chain credit history. Because Borrowers need to publicize their wallet address when proposing pools to Backers, their on-chain history becomes public to future creditors—even those off-chain. As more of global finance moves on-chain, creating a negative on-chain credit history is comparable for future credit opportunities to creating a negative off-chain credit score or record.

The second incentive is that Borrowers will likely want to continue borrowing from Goldfinch. The moment they are late on any payment, Borrowers are unable to borrow further from any Borrower Pool.

While not explicitly supported by the protocol, a third incentive is that Backers may form off-chain legal agreements with Borrowers, including requiring Borrowers to collateralize their on-chain loan with off-chain collateral. Backers may require such an agreement to be in effect, either with them directly or with another Backer, in order to be willing to supply capital. In these cases, the legal agreement and potential Investor recourse are another important incentive for Borrowers. Currently, all loans on Goldfinch are fully collateralized with off-chain assets via this mechanism.

In addition, while occasional defaults are common across finance, Backers will likely stop supplying more capital to any of a Borrower's pools if the Borrower is consistently late on repayments. It is up to Backer to evaluate whether Borrowers are qualified and have a good enough track record to garner Backer investment in any future pools the Borrower proposes.

Auditors

- ✓ Auditors perform human-level checks on Borrowers to confirm they are legitimate, helping to secure the protocol against fraud. Borrowers need the approval of Auditors to borrow from Borrower Pools. Currently, the protocol's Auditor system is not yet live, but it is expected that the community will introduce and vote on a proposal to implement auditing in the coming months.

Approval votes

Borrowers need an approval vote from Auditors in order to borrow. Auditors stake GFI in order to be selected for votes, and they earn GFI rewards when they vote with the majority of other Auditors, according to the rules described below.

- Anyone can be an Auditor by staking a minimum amount of GFI and passing the [Unique Entity Check](#). When a vote is requested, the protocol selects 9 Auditors on a random basis weighted by the amount of GFI they have staked.

- When selected for a vote, Auditors evaluate whether Borrowers appear to be legitimate.

In this vote, the Auditors are not evaluating the Borrower's creditworthiness — rather, they are providing a confirmation that the Borrower does what they claim to do and that they do not appear to be colluding with any other participants.

- Auditors can do whatever they like to decide how to vote. In practice, they may review off-chain documents provided by Borrowers and communicate with Borrowers directly

through channels such as forums, email, and video calls. This can all occur off-chain on

a variety of platforms. The protocol only needs the final vote and is agnostic to how Auditors arrive at their vote.

Approval vote requests

Borrowers can request an approval vote once their first Borrower Pool has reached at least 20% of its limit and they have staked enough GFI to reward Auditors for the vote. If more than 2 Auditors vote “No”, the Borrower's full GFI staked amount is slashed.

In addition to the Borrower making their first approval request, anyone can use GFI to pay for an approval request at any time. This is helpful if someone believes a prior approval vote had an incorrect result, or if someone believes the Borrower has started to act fraudulently and should lose their approval.

Approval vote outcomes

Once selected, auditors have 48 hours to provide a , , or vote. Their GFI is slashed if they:

- don't vote within the 48 hour window,
- vote when the majority vote , or
- vote when the majority vote .

 If they vote , there is no penalty but also no reward.

Based on the way Auditors vote, there are three potential outcomes:

1. **Full Approval:** This occurs when there are at least 6 votes and no more than 1 vote. The Borrower is approved to access capital, and the Senior Pool allocates capital to their Borrower Pools.
2. **Backer-Only Approval:** This occurs when there are at least 6 or votes, and no more than 1 vote. The Borrower is approved to access capital, but the Senior Pool does not allocate capital to their Borrower Pools.
3. **No Approval:** This occurs when there is more than 1 vote, or when there are not enough votes to meet the above approval thresholds. The Borrower is not approved to access any capital.

Summary of auditor incentives

Auditors are incentivized to participate and vote correctly in order to earn GFI rewards. Also, by staking GFI, they are both incentivized to participate as expected in order to avoid having their stake slashed and are also naturally aligned with the long-term success of the protocol.

Backers

✔ Backers optimize for yield and selection

Backers are Investors who supply USDC to individual Borrower Pools. Backers evaluate individual deals, and lend directly to specific Borrower Pools with first-loss capital.

General Overview

Backers look at Borrower Pools and Callable Deals as investment opportunities. They evaluate the information Borrowers provide, and decide if they want to supply capital (junior tranche capital for Borrower Pools; unitranche for Callable Deals).

To track the different amounts that different participants supply, Backers receive an NFT when they supply capital. The NFT tracks the amount that was supplied and how much of it has been redeemed. Backers can use their NFT to redeem their specific portion of the available repayments in the Pool or Deal, depending on the type of deal.

Supplying to Callable Deals

The Callable Deal loan structure gives Backers the right to “call back” their invested capital before the loan term has ended. Borrowers are required to return 100% of this “called capital” at the end of a “call period.”

Callable loans are an existing structure in traditional finance that provides liquidity to investors by giving them the right to “call back” their capital at regular intervals. Goldfinch's initial design of callable deals sets call periods to occur every three months; call requests must be submitted **at minimum 60 days** before the end of the call period (any calls that occur less than 60 days before the closest upcoming repayment date will be paid on the second closest upcoming repayment date).

Supplying to Borrower Pools

Backers evaluate the information Borrowers provide and decide if they want to supply first-loss capital (junior tranche) to fund a Borrower Pool.

The Senior Pool provides additional second-loss (senior tranche) capital to the Borrower Pool according to the [Leverage Model](#). To account for the lower risk of the senior tranche, 20% of the senior tranche's nominal interest is reallocated to the junior tranche. In addition, the protocol retains 10% of all interest payments as reserves, which are managed by the decentralized Governance.

As a result, the Senior Pool earns an effective interest rate equal to 70% of the nominal interest rate. Or, in terms of the nominal interest rate, i_n , protocol reserve allocation, p , and junior reallocation percent, j :

$$i_{senior} = i_n * (1 - p - j)$$

Accordingly, based on these same inputs and the leverage ratio, r , Backers receive an effective interest rate of:

$$i_{junior} = i_n * (1 - p + r * j)$$

For example: Consider a Borrower Pool with a 15% interest rate and 4.0X leverage ratio. If the Backers supply \$200K, the Senior Pool will allocate another \$800K. Assuming the Borrower borrows the full \$1M for one year, they will pay $\$1M * 15\% = \$150K$ in interest. Of that, the Senior Pool receives $0.15 * (1 - 0.1 - 0.2) = 10.5\%$ interest, or $\$800K * 0.105 = \$84K$. The Backers receive $0.15 * (1 - 0.1 + 4 * 0.2) = 25.5\%$ interest, or $\$200K * 0.255 = \$51K$. The remaining \$15K is the 10% protocol reserve allocation.

For a step-by-step on supplying to Borrower Pools, read the documentation's [Investor How-To](#) section.

Early Backer Rewards

It is easier to feel confident supplying to a Borrower Pool when a lot of other Backers have already vetted and supplied to it, and the Senior Pool is already adding leverage. It is riskier to be the first one in a Borrower Pool. To incentivize early Backers, the protocol provides an additional GFI reward to all Backers who contribute early on, with the reward amount decreasing for later Backers as the Borrower Pool reaches its limit.

The protocol assigns the reward when a Backer supplies, but the reward is not immediately claimable. The percent of the reward that is claimable is proportional to the percentage of the full expected repayment of principal plus interest that the Borrower successfully repays. This ensures the Backer only receives the early Backer reward after the Borrower Pool proves valuable to the protocol.

Staking on Backers

 Note: As of August 2022 staking on Backers is not yet live on Goldfinch.

In addition to evaluating individual Borrower Pools, Backers may also evaluate other Backers in order to give them leverage. Backers can do this by staking GFI directly on another Backer.

Based on the amount of GFI staked on a given Backer, the Senior Pool uses the Leverage Model to calculate a leverage ratio and allocate capital whenever that Backer supplies to Borrower Pools. For example, if a Backer has a leverage ratio of 4.0X based on the GFI staked on them by other Backers, then anytime they supply to a Borrower Pool, the Senior Pool will allocate 4.0X of that amount.

The Senior Pool provides this leverage up to a maximum total that is calculated as the leverage ratio multiplied by the total value of GFI staked on that Backer. For example, if the Backer has \$1M worth of GFI staked on them with a 4.0X leverage ratio, the Senior Pool will allocate up to \$4M total leverage.

When GFI is staked on a Backer, that GFI serves as collateral against potential defaults for that Backer's positions in Borrower Pools. When a Borrower defaults, the GFI staked on all the Backers in that pool are reallocated to the senior tranche until the senior tranche is made whole on their expected payments. This incentivizes Backers to stake on other Backers who supply to safe Borrower Pools.

To reward Backers for staking GFI on other Backers, the protocol distributes GFI to them on a regular basis. The protocol allocates the distributions in proportion to the interest their leveraged GFI earns. This incentivizes Backers to stake on other Backers who supply to high-yielding Borrower Pools.

Summary of Backer mechanics

Backers have an incentive to provide first-loss capital to Borrower Pools as they can receive both early Backer rewards and higher effective yields based on the Senior Pool leverage. In the future they will also have an incentive to stake GFI on other Backers as they will be able to earn additional rewards when that Backer supplies to Borrower Pools.

Liquidity Providers

✔ Liquidity Providers optimize for diversification and liquidity

Liquidity Providers are Investors who supply USDC to Goldfinch by investing in the Senior Pool. This Senior Pool capital is automatically allocated across Borrower Pools.

Supplying to the Senior Pool

Liquidity Providers supply capital to the Senior Pool in order to earn yields optimized for ease and diversification. The Senior Pool automatically allocates that capital across the senior tranches (second-loss) of Borrower Pools according to the [Leverage Model](#).

In this way, when more Backers actively evaluate a Pool's terms and decide to invest first-loss capital in it, the Senior Pool will in turn automatically allocate more second-loss capital to the Pool. The Senior Pool thereby provides both diversification across Borrower Pools and seniority to the first-loss capital of Backers. This process does not involve seeking the permission of different Borrowers.

To compensate Backers for providing the work of both evaluating Borrowers Pools and providing first-loss capital, 20% of the Senior Pool's nominal interest is reallocated to Backers.

As a result, the Senior Pool earns an effective interest rate equal to 70% of the nominal interest rate. Or, in terms of the nominal interest rate, i_n , protocol reserve allocation, p , and junior reallocation percent, j :

$$i_{senior} = i_n * (1 - p - j)$$

Accordingly, based on these same inputs and the leverage ratio, r , Backers receive an effective interest rate of:

$$i_{junior} = i_n * (1 - p + r * j)$$

For example: Consider a Borrower Pool with a 15% interest rate and 4.0X leverage ratio. If the Backers supply $\$200K$, the Senior Pool will allocate another $\$800K$. Assuming the Borrower borrows the full $\$1M$ for one year, they will pay $\$1M \times 15\% = \$150K$ in interest. Of that, the Senior Pool receives $0.15 \times (1 - 0.1 - 0.2) = 10.5\%$ interest, or $\$800K \times 0.105 = \$84K$. The Backers receive $0.15 \times (1 - 0.1 + 4 \times 0.2) = 25.5\%$ interest, or $\$200K \times 0.255 = \$51K$. The remaining $\$15K$ is the 10% protocol reserve allocation.

For a step-by-step on supplying to the Senior Pool, read the documentation's [Investor How-To](#) section.

FIDU

When Liquidity Providers supply to the Senior Pool, they receive an equivalent amount of FIDU. FIDU is an ERC20 token.

At any time, Liquidity Providers can withdraw their position by depositing their FIDU to a [Withdrawal Request](#), to redeem their FIDU for USDC at an exchange rate based on the net asset value of the Senior Pool, minus a 0.5% withdrawal fee. This exchange rate for FIDU increases over time as interest payments are made back to the Senior Pool.

Withdrawal Requests are fulfilled every two weeks. It is possible that when a Liquidity Provider wants to withdraw, the Senior Pool may not have sufficient USDC because it has been borrowed by Borrowers. If there is enough USDC unutilized in the Pool to honor all outstanding withdrawal requests at the end of the distribution period, all withdrawers will receive 100% of the value of the FIDU they requested to withdraw, in USDC, minus the 0.5% withdraw fee.

If there is not enough USDC in the Senior Pool to honor all outstanding Withdrawal Requests at the end of the period, due to the USDC being actively utilized by Borrowers, all available USDC will be allocated to withdrawers pro-rata

You can learn more about how Withdrawal Requests work in the [Liquidity section](#) of the documentation.

Summary of Liquidity Provider mechanics

Liquidity providers are incentivized to supply to the Senior Pool in order to earn automatically allocated yields, as their capital is diversified across different Borrower Pools.

Leverage Model

- ✓ The Leverage Model determines how much capital the Senior Pool allocates toward each Borrower Pool, based on how much it `trusts` each Borrower Pool. Currently, the automated Leverage Model as described here is not yet live, but it was included in the community-approved roadmap for near-term development linked [here](#) ↗.

Trust Through Consensus

In order to determine how to allocate capital from the Senior Pool, the protocol uses the novel principle of "trust through consensus." This means that while the protocol doesn't trust any individual Backer or Auditor, it does trust the collective actions of many of them. At a high level: when more Backers supply capital to a given Borrower Pool, the Senior Pool increases the ratio with which it adds leverage.

Because this approach relies on counting individual Backers, the protocol must ensure they are in fact represented by different people. Therefore, all Backers, Borrowers, and Auditors require a `unique entity check` to participate (see the [Unique Entity Check section](#)).

Leverage Model Formula

The leverage amount, A , that the Senior Pool allocates is determined by the formula, $A = S * D * L$ where:

- S is the total capital supplied by Backers.
- D is the distribution adjustment on a scale of 0 to 1, which accounts for how evenly distributed the Backers are. D is closer to 0 when the distribution is skewed and closer to 1 when the Backers are more equally distributed. This ensures no single Backer has an outsized influence. The formula for D uses the percent supplied by each Backer, s_n , and is based on the Herfindahl-Hirschman Index:

$$D = 1 - \sum_{i=1}^n s_n^2$$

- L is the leverage ratio on a scale of 0 to the maximum potential leverage ratio. Based on the number of Backers, b , the leverage ratio increases linearly from B_{min} , the minimum number of Backers necessary for leverage, to B_{max} , the maximum number of Backers necessary to achieve the maximum potential leverage, L_{max} :

$$L = L_{max} * \frac{\max(0, b - B_{min})}{B_{max} - B_{min}}$$

Membership

Join a Membership Vault as an active Goldfinch Investor to upgrade your Goldfinch participation and support Goldfinch's growth and expansion.

Introduction to Goldfinch Membership

- Goldfinch Membership is the first phase of a broader tokenomics redesign ([Tokenomics v2 ↗](#)), which was approved by the community in 2022 and focuses on enhancing the utility of GFI.
- Membership is designed to empower Goldfinch Investors to support the protocol's growth while increasing their participation.

Goldfinch Members receive yield enhancements via Member Rewards, which have been earmarked from the Goldfinch treasury and are distributed pro-rata based on one's Membership Vault position. In addition, Members will gain access to exclusive communication channels, special offers, and more.

Encouraging Investors to hold GFI will help to increase community participation in [governance ↗](#)—which requires GFI—as well as further aligning and incentivizing the Goldfinch community's interest in the long-term success of the protocol.

Membership also encourages the single biggest lever for network growth, increased TVL, resulting in a more robust and secure protocol ecosystem while also rewarding the global community of Goldfinch participants who continue to contribute to Goldfinch's growth and resilience.

Becoming a Goldfinch Member

- There are only two requirements to participate in Goldfinch Membership: being a [Senior Pool LP](#) or [Backer](#) on Goldfinch (represented by holding staked FIDU and/or an active Backer NFT), and holding [GFI](#).
- To become a Member, one only needs to lock their staked FIDU or Backer NFTs, plus GFI, into the [Membership Vault in the Goldfinch dapp ↗](#).

- To optimize the yields you can receive, balanced is best: matching the dollar value of the GFI and assets you deposit to the vault will balance the vault ratio and increase your potential Member Reward yield.

You can withdraw your deposited assets from the vault at any time, and there is no risk of slashing when participating in a Membership Vault. However, if you withdraw your assets from the vault before the end of a reward cycle (weekly), you will forfeit any Member Rewards accumulated during that cycle.

While future releases are expected to launch support for joining a Membership Vault by depositing Curve FIDU/USDC LP Tokens or staked Curve FIDU/USDC LP Tokens, at launch the vaults only accept Backer PoolToken NFTs, representing participation in a Goldfinch Borrower Pool, or staked FIDU, Goldfinch Senior Pool tokens that have been staked in the dapp to receive extra GFI rewards. If you hold unstaked FIDU, you can stake it to be able to supply it to a Membership Vault by following [this guide](#).

Member Rewards

- Member Rewards are a form of yield enhancement to further align and incentivize Goldfinch Investors' interest in the long-term success of the protocol.
- Member Rewards are distributed weekly in [FIDU](#).

Member Rewards come from an earmarked percentage of Goldfinch's Treasury, and are paid pro-rata to Members—meaning that the higher the percentage of the Membership Vault's assets one supplied during a reward cycle, the higher percentage of Member Rewards one will receive for that cycle.

Member Rewards are distributed weekly in FIDU, naturally increasing Members' exposure to Goldfinch's Senior Pool to increase their opportunities to receive yields. Rewards must be claimed manually, can then be staked to receive additional GFI rewards, and then can be deposited to the Membership Vault alongside GFI, increasing one's interest in the Membership Vault to receive additional Member Rewards.

If a participant deposits assets into the Membership Vault during a weekly reward cycle their assets will begin accumulating Member Rewards at the beginning of the next weekly reward cycle. While Members can withdraw their deposited assets at any time to exit Membership, if they withdraw before the end of a weekly reward cycle they will forfeit any Member Rewards they would have received during that cycle.

Following launch, Members will also have access to exclusive communication channels, special offers, and more.

Member Rewards calculation

The Member Rewards share calculation is a governance-controlled parameter that determines the optimal share of Capital and GFI for maximizing rewards in Membership Vaults.

A Member's precise share of Member Rewards is calculated using the [Cobb-Douglas function ↗](#), an equation commonly used in economic analysis for balanced input relationships generating an output, and also used by protocols such as 0x and The Graph for reward systems.

All values are for a given weekly period:

The two variables in the Member Reward equation that are determined by governance are:

- μ = Share of earmarked treasury used for Member Rewards
 - You can think of this as "What percentage of the earmarked treasury allocation should be allocated to members?"
 - Currently, $\mu = 50\%$
- α = Capital supply amplification weight.
 - You can think of this as "How much should we weight locked capital supply, versus locked GFI for fee shares?"
 - Currently $\alpha = 0.50$

The μ and α values can be changed by governance vote. $\mu = 50\%$ allows the protocol to build sufficient stablecoin reserves for future needs, like community grants, and $\alpha = 0.50$ is a good starting point to observe how dynamics play out between the balances of capital supply and GFI.

You can learn in-depth about how Member Rewards are calculated in the original [Membership Governance proposal \(GIP-13\)](#) ↗.

How are the estimated Member Rewards in the app calculated?

The estimated Member Rewards displayed in the Goldfinch platform are calculated based on a) one's Membership Vault position, and b) on the distribution of total capital in the Membership Vault overall.

- This number is an estimate because rewards are distributed pro-rata to Members at the end of each reward cycle—as such, other Investors entering or exiting the Membership Vault during a reward cycle will change the percentage of rewards one is expected to receive, due to changing what percentage of the Vault one's individual position represents.
- The estimated Member Rewards displayed is a dynamic number that reflects the distribution of total capital in the Membership Vault overall at that moment in time and one's Member Vault position at that moment in time.

Investor Incentives

All investors on Goldfinch receive two core incentives for their participation in the protocol:

- **USDC APY** — The base-level USDC return an Investor receives for participating in Goldfinch, generated from Borrowers' interest payments on their loans. The USDC APY is defined in the Borrowers' financing terms when they establish Borrower Pools. For Backers, this APY is a fixed rate as defined by the Borrower Pool's terms. For Liquidity Providers, this rate is estimated as the Senior Pool's USDC returns vary based on the Senior Pool's usage and balance.
- **Investor Rewards** — The additional GFI return all Investors on Goldfinch receive in exchange for their participation in the protocol. Investor Rewards are an estimated return as they are not defined by Borrowers' financing terms, but are instead dependent on tokenomics and network dynamics.

Investor Rewards vary based on whether one is participating in Goldfinch as a Backer or as a Liquidity Provider. There are specific rewards to incentivize Backers for taking on the risk of providing the protocol's junior capital as well as for their work evaluating Borrower Pool deals, while Liquidity Providers can receive Investor Rewards by participating in Senior Pool Liquidity Mining. Read on for the breakdown of Backer Incentives and Senior Pool Liquidity Mining.

In addition, LPs of the Curve FIDU<>USDC Liquidity Pool can also earn rewards for staking their Curve LP tokens. Learn more about Curve LP incentives in the [Staking section](#) of the documentation.

Backer Incentives

The total APY (Annual Percentage Yield) earned by Backers of a Borrower Pool is the sum of three parts:

- The base interest **USDC APY** of a Pool's Junior Tranche is the base-level incentive a Backer can receive from participating in the protocol. This base USDC APY originates from the Borrower's repayments and is a core feature of the Goldfinch protocol.
- **Backer Bonus** is the APY from GFI earned uniquely by Backers, which is made up of the combined BackerRewards contract functions of **Backer Rewards**, from the Pool's interest repayments, and **Backer Staking Rewards**, the APY from GFI earned by Backers equivalent to the APY from GFI earned by Liquidity Providers who supply to the Senior Pool.

In addition to the total APY earned by backers:

- **Early Backer Airdrop** — A GFI reward provided to Backers who contributed to a Pool before the implementation of the community governance proposals for Backer Rewards and Backer Staking Rewards mechanisms.

Backer Rewards and Backer Staking Rewards are functions of the BackerRewards contract. This was passed by community governance to ensure that a Backer's incentives are always higher than those of a Senior Pool participant's, in order to preserve the protocol's consistent risk/reward tradeoff, with Backers, who take on the most risk, rewarded the most highly.

The Early Backer Airdrop was passed by [community governance ↗](#) to retroactively reward Backers of pools that were funded prior to the development of the Backer Rewards and Backer Staking Rewards mechanisms.

Staking on Backers, detailed below, is an element of the [Goldfinch Whitepaper ↗](#) and as of April 2022 have not yet been implemented as a live feature on the protocol.

Early Backer airdrop

Backers of pools that were funded before the Backer Rewards and Backer Staking Rewards mechanisms were developed received a one-time airdrop, to reward their early support of the protocol. The parameters of the airdrop were defined in the [proposal](#)  passed by community governance.

Backer rewards

Backer Rewards are distributed as GFI rewards to Backers for interest payments made by different Borrowers. For every dollar of interest that a Borrower repays, a Backer will earn an amount of GFI. To incentivize early participation in the protocol, this amount decreases based on the total amount of interest that has been repaid to the Goldfinch protocol as a whole.

The protocol does this to incentivize Backers who are evaluating pools to fully take into account the likelihood that interest payments will be made successfully and on time to the Pool, and to provide an incentive for Backers to hold Borrowers accountable to their obligations.

This means that how much GFI a Backer earns for a dollar of interest is not a fixed value, i.e. at one point in time a Backer may earn a large amount of GFI per dollar of interest, but later in the protocol's lifecycle they may earn less GFI per dollar of interest. Concretely, the rewards fall off on a square root curve and will eventually fall to 0 once \$100MM total dollars of interest has been repaid by Borrowers to the Goldfinch protocol as a whole. As such, Backers who participate in the protocol's earliest pools will receive more rewards than Backers who take part later on, when the protocol's usage has grown.

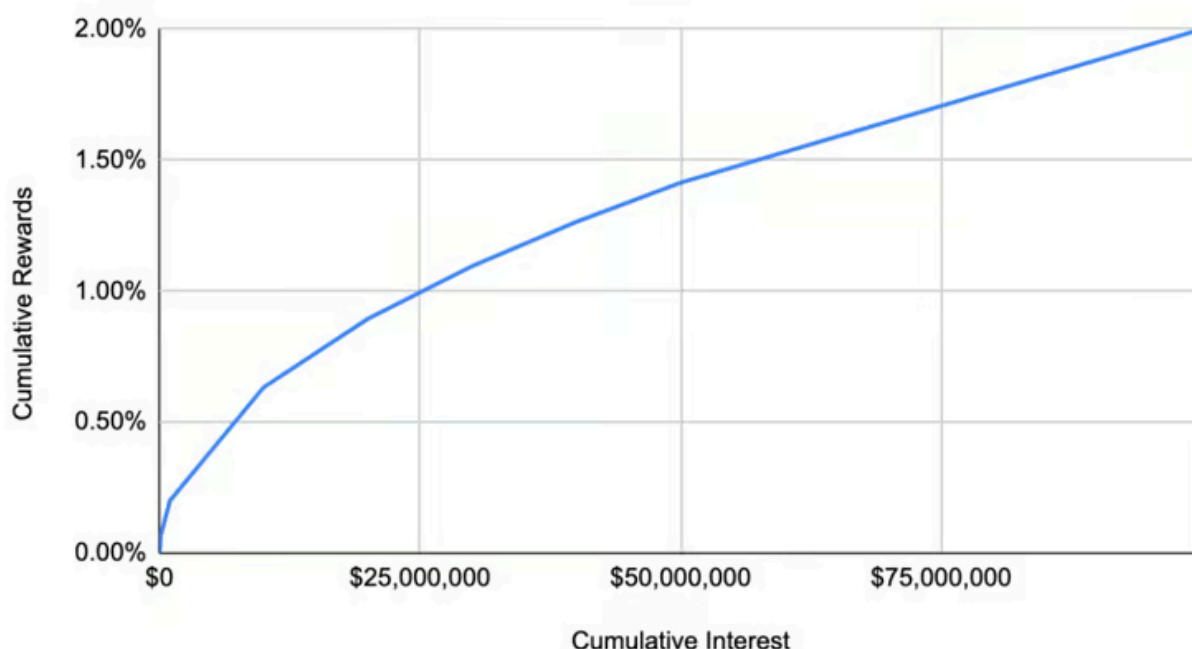
Backer staking rewards

Backer Staking Rewards take the form of GFI distributed to Backers in exchange for their role taking on the additional risk of providing first-loss capital to Backer Pools. This is achieved by providing Backers equivalent GFI rewards as if they had deposited and staked in the protocol's Senior Pool. The protocol does this so that Backers are compensated fairly for the increased risk they take on in the protocol.

Upon a successful repayment from a Borrower, a Backer will be able to claim the full amount of Backer Staking Rewards they've earned since the last time they claimed. Partial repayments made by the Borrower to their Pool do not entitle Backers to rewards, only full payments. However, if a Borrower eventually does make a repayment on their outstanding obligations, Backers will not be penalized and will receive the same rewards as if the incomplete payment never occurred.

Backer Staking Rewards is an implementation of the accepted Goldfinch governance proposal titled "[Backer Participation in Staking Rewards ↗](#)."

Cumulative Rewards vs. Cumulative Interest



BackerRewards contract technical design

Backer Rewards Interest repayments to a Pool earn GFI for Backers according to the parameters of the approved [backer liquidity mining proposal ↗](#). Thus 2% of the total GFI supply is allocated for this purpose, to be earned upon the first \$100M of interest repaid to the protocol. GFI are earned at an "exponentially decelerating rate" across all interest dollars repaid to eligible pools; the exact formula can be found in the documentation of the smart contract.

Backer staking rewards

From the time of drawdown of the Pool's Backer capital, until the term end time of the loan, the protocol calculates an amount of "virtual" FIDU, corresponding to the borrowed amount of capital supplied by the backer, that would have earned the APY-from-GFI of Senior Pool staking, if that amount of FIDU had been staked in the Senior Pool. The backer earns an equivalent^[^1] amount of GFI as they would have earned from staking this amount of "virtual" FIDU in the Senior Pool.

[^1]: We say "equivalent" rather than "equal" due to a subtlety of implementation in the calculation logic relating to the term end time of the loan.

Distribution parameters

The GFI earned from Backer Rewards and Backer Staking Rewards is not subject to a lockup period once it is earned. But the GFI of Backer Rewards and Backer Staking Rewards is only earned by Backers upon an interest repayment to the pool by the Borrower, at which time the earnings from Backer Rewards and Backer Staking Rewards are checkpointed and a corresponding additional amount of GFI since the last checkpoint becomes claimable by backers.

Staking on Backers

Staking on Backers, detailed below, is an element of the [Goldfinch Whitepaper](#) ↗ which as of June 2022 has not yet been implemented as a live feature on the protocol.

In addition to evaluating individual Borrower Pools, Backers may also evaluate other Backers in order to give them leverage. Backers can do this by staking GFI directly on another Backer. Based on the amount of GFI staked on a given Backer, the Senior Pool uses the Leverage Model to calculate a leverage ratio and allocate capital whenever that Backer supplies to Borrower Pools. For example, if a Backer has a leverage ratio of 4.0X based on who has staked GFI on them, then anytime they supply to a Borrower Pool, the Senior Pool will allocate up to 4.0X of that amount.

The Senior Pool provides this leverage up to a maximum total that is calculated as the leverage ratio multiplied by the total value of GFI staked on that Backer. For example, if the Backer has \$1M worth of GFI staked on them with a 4.0X leverage ratio, the Senior Pool will allocate up to \$4M total leverage.

When GFI is staked on a Backer, that GFI serves as collateral against potential defaults for that Backer's positions in Borrower Pools. When a Borrower defaults, the GFI staked on all the Backers in that pool are reallocated to the senior tranche until the senior tranche is made whole on their expected payments. This incentivizes Backers to stake on other Backers who supply to safe Borrower Pools.

To reward Backers for staking GFI on other Backers, the protocol distributes GFI to them on a regular basis. The protocol allocates the distributions in proportion to the interest their leveraged GFI earns. This incentivizes Backers to stake on other Backers who supply to high-yielding Borrower Pools.

Summary of Backer incentives

Backers have an incentive to provide first-loss capital to Borrower Pools because they can receive both Backer rewards and higher effective yields based on the Senior Pool leverage. To help ensure that Backers are always receiving as much or higher of an APY on their pool participation as Senior Pool Participants, Backer Rewards and Backer Staking Rewards are also distributed based on participation. In the future, Backers will also have an incentive to stake GFI on other Backers as they will be able to earn additional rewards when that Backer supplies to Borrower Pools.

Senior Pool Liquidity Mining

Liquidity Providers can earn GFI by staking the FIDU they receive for supplying to the Senior Pool.

GFI tokens are granted at a variable distribution rate, which is based on a target pool balance set by Governance. More tokens are distributed when the pool is under the target, and less are distributed when it is over the target. The further under the target the pool balance is, the more GFI tokens are distributed. This helps incentivize a healthy utilization and APY for the pool, relative to loans outstanding.

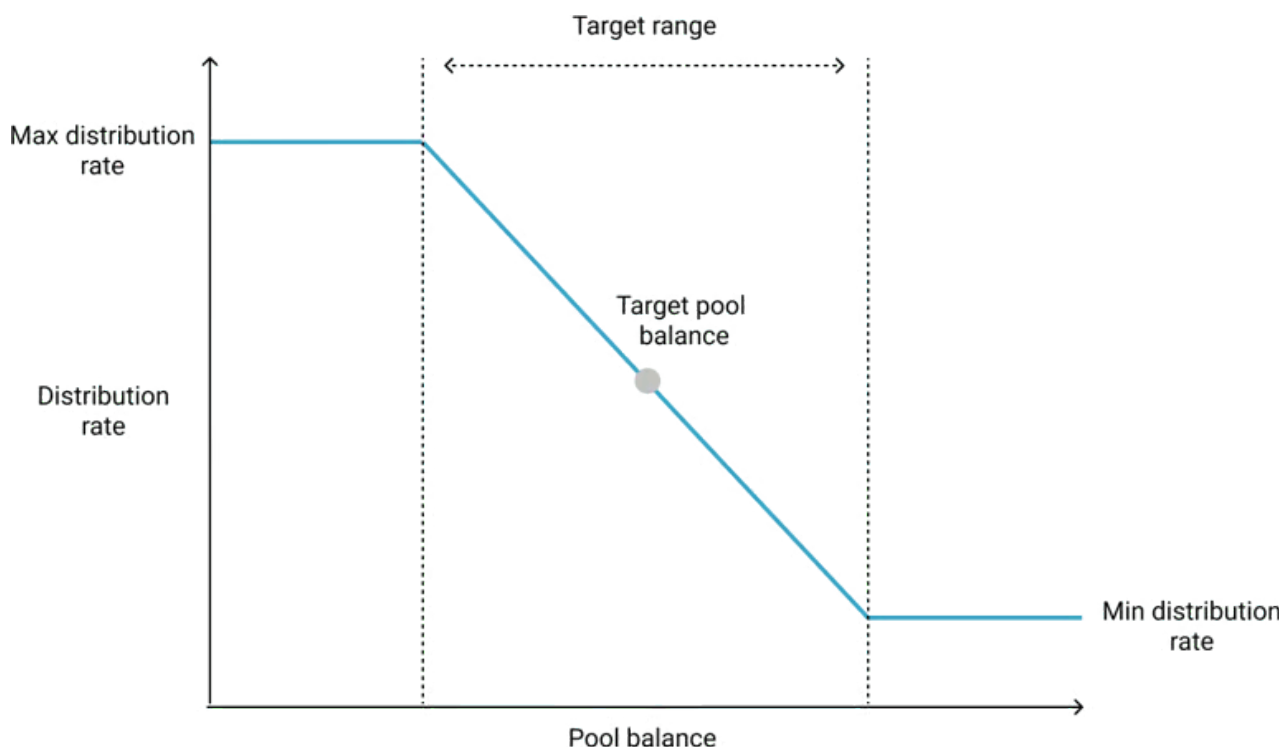
All FIDU liquidity miners, including those who staked FIDU prior to the implementation of community proposal GIP-10 on July 20, 2022, can withdraw from staking at any time and receive the full value of their vested GFI liquidity mining rewards at withdrawal. See below for more details.

Distribution rate

The GFI distributions rate is calculated using several parameters, all set by Governance:

- **Target balance:** The ideal total balance of the Senior Pool. The community can decide this balance based on expected future capital needs. It should be high enough to attract the amount of capital the protocol will need, but not too high that the protocol unnecessarily distributes GFI to unused capital.
- **Minimum rate:** The lowest possible rate of GFI distributed per second.
- **Maximum rate:** The highest possible rate of GFI distributed per second.
- **Target range:** The range around the "Target balance" along which the min and max distribution rate are applied. It is represented as two percentages (e.g. min: 50% of the target balance, max: 200% of the target balance).

The reward rate can be thought of as a piecewise linear function that looks something like this:



- Inside the target range, the distribution rate linearly decreases from "Maximum rate" to "Minimum rate"
- Below the target range, the distribution rate is constant at "Maximum rate"
- Above the target range, the distribution rate is constant at "Minimum rate"

In other words, when the pool balance is below its target, the distribution rate will be higher, incentivizing more people to supply capital to the pool. When the pool balance is above its target, the distribution rate will be lower, incentivizing withdrawals.

Current parameters

As of January 11 2021, the distribution rate parameters have been set as follows:

- **Target balance:** \$100M
- **Minimum rate:** 0 GFI
- **Maximum rate:** 0.5% of GFI supply per month (this equals 0.217438574961948 GFI / second)
- **Target range:** \$50M to \$200M (50% to 200%)

Governance can always decide to change these parameters.

Unlock schedule

The GFI distributions received from liquidity mining can be withdrawn at any time, with no unlock period or slashing. FIDU stakers can stake and unstake at will without forfeiting any of the GFI rewards they have earned. You can learn how to stake FIDU in the [Participating in Liquidity Mining](#) section of the documentation.

FIDU staked prior to July 20, 2022

Before July 20, 2022, the GFI distributions received from liquidity mining unlocked linearly over the first 12 months of staking with early withdrawals forfeiting any remaining locked rewards. This meant that a liquidity provider could withdraw from staking whenever they chose, but would forfeit some GFI distributions if they withdrew before 12 months.

This was updated by the implementation of community proposal GIP-10 to make the GFI rewards received for liquidity mining withdrawable at any time with no unlock period or slashing. To ensure that the new system did not hurt existing FIDU stakers in any way when implemented, the proposal also removes slashing for Liquidity Providers who staked FIDU before July 20, 2022.

If a Liquidity Provider who staked FIDU before July 20, 2022 withdraws, the part of their GFI rewards that would have been slashed under the original rewards design will instead not be slashed and will unlock over their remaining 12 month period. The Liquidity Provider will continue to earn their remaining GFI rewards, and can un-stake and re-stake their FIDU to begin receiving GFI rewards with no unlock/vesting period moving forward.

Unique Entity Check

Because the Leverage Model relies on "trust through consensus", and also to help avoid [Sybil Attacks ↗](#), it is critical for the protocol to have confidence that each Borrower, Backer, and Auditor is a unique entity. Therefore, they must each be verified with a `Unique Entity Check` before they can participate.

Currently, the protocol uses [Unique Identity \(UID\)](#) as its Unique Entity Check provider. View the [Unique Identity \(UID\)](#) section of the documentation to learn about UID, how it works, and how to get started.

Discussion of Fraud Resistance

Because the protocol does not require crypto overcollateralization, new potential vectors for fraud are possible. It is worth discussing each one in depth, and how the protocol builds resistance against it. Please note that these scenarios focus on malicious or dishonest activity, not poor performance of well-intentioned borrowing.

Fraudulent Borrower, honest Backers

A fraudulent Borrower could first attempt to fool both Auditors and Backers into thinking they are legitimate, and then borrow capital without repaying it.

The first guard against this are the Auditors, who must approve Borrowers before borrowing. Because Auditors are randomly selected, it is difficult to collude with them.

The second guard are the Backers, who are highly incentivized to analyze their investment choices closely as they supply higher-risk junior capital—meaning they will be the last to be repaid if a Borrower was to default. It is likely that Backers will want to do extra research on Borrowers and potentially communicate with them directly.

Lastly, Backers may sign off-chain legal contracts with Borrowers, which opens Borrowers to legal recourse. Currently all loans on Goldfinch are fully collateralized with off-chain assets and legal contracts.

Borrower collusion with Backers

A Borrower could collude with people they know to act as Backers and supply to their Borrower Pool. This would artificially increase the leverage ratio and fool the Senior Pool into allocating additional capital.

The first guard against this are the Auditors, who must approve Borrowers before they are able to borrow. Because Auditors are randomly selected, it is difficult to collude with them.

The second guard is that it requires many individually verified Backers to supply significant amounts of upfront capital in order for the Senior Pool to provide leverage, which makes such collusion with Backers difficult and expensive.

Lastly, the Unique Entity Check adds Sybil Resistance by making it difficult to programmatically create fake Backers.

Borrower collusion with Auditors

A Borrower could collude with Auditors to obtain approval for creating Borrower Pools when the Borrower is in fact not legitimate.

The first guard against this is that the requirement of a Unique Entity Check prevents a Sybil Attack, where fake Auditors are programmatically created to overwhelm the system. Instead, each Auditor must be a verified entity.

The second guard is that Auditors must stake GFI in order to participate, which is slashed if they vote differently than the majority of Auditors. This incentivizes Auditors to act honestly in order to preserve their stake.

The third guard is that Auditors are randomly selected, weighted by their staked GFI, so it would require staking a significant amount of upfront capital to be chosen frequently enough to skew the votes.

The fourth guard is that anyone can request an approval at any time, so it would require colluding for all potential future votes rather than just one.

Lastly, even if a fraudulent borrower successfully colludes with Auditors, they must also convince many Backers to risk their own capital by depositing to the Pool.

Fraudulent Backers, honest Borrowers

An individual or group of Backers might supply to a particular Borrower Pool even when they don't view it as a good risk. This would artificially increase the leverage ratio and fool the Senior Pool into allocating additional capital, boosting the Backers' returns.

The first guard against this is that the Unique Entity Check requires each Backer to be verified, preventing a Sybil Attack of programmatically-created Backers and instead requiring the coordination of many people to achieve Backer collusion.

The second guard against this is that it requires the Backers to take real risk by supplying first-loss capital. The Backers only achieve higher returns if the Borrower does in fact pay back what they borrow, in which case it is beneficial to all participants in the protocol, including the Senior Pool.

Staking

There are currently two types of staking accessible on the Goldfinch protocol. All staking and participation incentives are driven and established via [Goldfinch's community governance](#) process. Staking can be accessed in the Goldfinch protocol application at app.goldfinch.finance/stake ↗.

Staking FIDU-USDC Curve LP Positions

Following a successful [community governance proposal](#) ↗, Liquidity Provider (LP) positions from participation in the Curve FIDU-USDC Pool can be staked on Goldfinch to receive additional [GFI](#) liquidity mining rewards.

GFI rewards for staking Curve LP positions come from the existing distribution parameters of the Senior Pool Liquidity Mining program, including being subject to the same unlock schedule as well.

Participants can deposit their unstaked FIDU or USDC into the Curve FIDU-USDC liquidity pool by visiting the [Stake](#) ↗ tab on the Goldfinch protocol platform, and selecting either Deposit FIDU or Deposit USDC under LP on Curve. In the same interface, a participant can select the check box to simultaneously stake their resulting Curve LP positions on the Goldfinch platform in order to receive GFI rewards for their participation.

In addition, those participating in Senior Pool Liquidity Mining by staking FIDU on Goldfinch can also deposit their staked FIDU into the Curve FIDU-USDC liquidity pool via the Goldfinch protocol platform, to earn additional LP rewards without needing to unstake from Goldfinch. Participants can migrate their staked FIDU to deposit into the Curve FIDU-USDC pool by visiting the [Stake](#) ↗ tab in the platform, and selecting FIDU, then Migrate, under Stake on Goldfinch.

Current parameters

As of January 11 2021, the distribution rate parameters have been set as follows:

- **Target balance:** \$100M

- **Minimum rate:** 0 GFI
- **Maximum rate:** 0.5% of GFI supply per month (this equals 0.217438574961948 GFI / second)
- **Target range:** \$50M to \$200M (50% to 200%)

Governance can always decide to change these parameters.

Unlock schedule

To encourage long-term participation, the GFI distributions received from liquidity mining unlock linearly over the first 12 months of staking. This means that while you can withdraw whenever you like, if you withdraw before 12 months, you will forfeit some of your GFI distributions. For example, after 3 months you'd forfeit 75%, after 6 months you'd forfeit 50%, and so on. After 12 months, you will continue to receive distributions, and will never forfeit any GFI. See below for more details.

You can read Goldfinch's documentation on [Senior Pool Liquidity Mining](#) to learn more about the Senior Pool Liquidity Mining's GFI distribution rate, parameters, and unlock schedule, which are the same parameters used for Staking FIDU-USDC Curve LP Positions.

Technical specifications

Implementing the community governance proposal to allow staking FIDU-USDC Curve LP positions for GFI involved augmenting Goldfinch's [StakingRewards contract](#) ↗ to accept FIDU-USDC Curve LP tokens and calculate rewards based on the user's Curve LP position.

The modifications of the `StakingRewards` contract could enable other types of staked positions on the protocol in the future, should the community pass further proposals to do so. Currently, the `StakingRewards` contract only supports staking FIDU directly, and as such the contract calculates individual rewards based on the amount of FIDU staked. In order to support any potential future community desire for staking tokens in the future, an additional calculation is performed at the time of staking to get the exchange rate used to convert Curve LP tokens to an effective FIDU amount. The exchange rate for a Curve LP position will be calculated using the [virtual price](#) ↗ of the LP token given by Curve.

Note: The Goldfinch protocol uses Curve's virtual price to calculate the effective FIDU amount instead of spot prices or live ratios of the FIDU-USDC Curve pool to prevent flash loan attacks.

Senior Pool Liquidity Mining

Participants can earn [GFI](#) by staking the FIDU received from supplying capital (USDC) to the [Senior Pool](#). Read our documentation on [Senior Pool Liquidity Mining](#) to learn more about the Senior Pool Liquidity Mining's GFI distribution rate, parameters, and unlock schedule.

Participants can stake their FIDU for GFI by visiting the [Stake ↗](#) tab in the platform, and selecting FIDU under Stake on Goldfinch.

Using the Goldfinch protocol application, staked FIDU can also be deposited in the Curve FIDU-USDC pool to earn additional LP rewards without needing to unstake from Goldfinch. To learn more about the community-driven incentives for staking FIDU-USDC Curve LP positions on Goldfinch, read the [Staking FIDU-USDC Curve LP Positions](#) section above.

Participants can migrate their staked FIDU to deposit into the Curve FIDU-USDC pool by visiting the [Stake ↗](#) tab in the platform, and selecting FIDU, then Migrate, under Stake on Goldfinch.

Backer Bonus incentives

To ensure that [Backers](#), who do not receive FIDU in exchange for providing capital to Borrower Pools, are compensated fairly for their participation in the protocol, Backer Staking Rewards are GFI rewards earned by Backers equivalent to the APY from GFI earned by liquidity providers who supply to the Senior Pool and stake their FIDU. Read our documentation on [Backer Incentives](#) to learn more about how Backers receive GFI rewards.

GFI Token

GFI is the official token of Goldfinch.

 The [contract address](#) for GFI is `0xdab396cCF3d84Cf2D07C4454e10C8A6F5b008D2b.`

The Goldfinch Protocol's core native token is GFI, and it is used for the following purposes:

- *Community Governance:* GFI is the token used for Community Governance. GFI holders can participate in governance to decide the direction of the protocol. [Learn more about Governance here.](#)
- *Participant Incentives:* All participants receive ongoing rewards to incentivize their participation. This includes liquidity mining for Liquidity Providers who supply to the Senior Pool. Learn more about Investor Incentives [here](#), and Staking [here](#).
- *Community Grants:* The community can provide grants and/or bounties to participants that meaningfully contribute to the Goldfinch protocol and ecosystem.

According to the whitepaper, GFI will also serve the following functions. These are not live yet, but it is expected that the community will introduce and vote on proposals for them in the coming months:

- *(More) Participant Incentives:* Backers who both supply to Borrower Pools and stake on other Backers, Auditors who stake to participate in votes, and Borrowers who successfully repay their pools.
- *Backer Staking:* Backers can stake their GFI tokens on other Backers in order to give them additional leverage when participating in Borrower Pools. This GFI also serves as a backstop against potential loan defaults.
- *Auditor Votes:* Auditor votes are required to grant Borrowers permission to borrow from the protocol, and Borrowers pay for these votes with the GFI token.
- *Auditor Staking:* Auditors stake the GFI token in order to be selected to participate in Auditor Votes.

You can learn more about the GFI distribution parameters and inflation in the [Tokenomics section](#) of the documentation.

Liquidity & LP Withdrawals

Senior Pool LPs: FIDU Liquidity via Withdrawal Requests

- As outlined in [GIP-25 ↗](#), LPs must now submit a Withdrawal Request to withdraw FIDU from the Senior Pool. There is a 0.5% withdrawal fee for redeeming FIDU for USDC.
- A single Withdrawal Request may be fulfilled over multiple distribution periods.
- Distributions happen every 2 weeks, and distribution amounts are variable based on availability of capital in the Senior Pool and total amount requested.
- A Withdrawal Request remains active until it is completely fulfilled or canceled. Once a Withdrawal Request has been made, it can only be increased or canceled, not reduced.
- If a Request is canceled before it is fully fulfilled, the LP is charged a cancellation fee of 1% of the total request.

At any time, Senior Pool [Liquidity Providers ↗](#) with a [Unique Identity NFT \(UID\) ↗](#) may submit a request to withdraw USDC from the Senior Pool by depositing their FIDU into a Withdrawal Request. There is a 0.5% withdrawal fee for redeeming FIDU for USDC on Goldfinch, contributing funding to the Goldfinch treasury.

FIDU that is staked or supplied to a Membership Vault must be unstaked or removed from the Vault before an LP can submit a request to withdraw it from the Senior Pool.

Various Goldfinch community members and third-parties have taken steps to create additional mechanisms to withdraw capital from the Senior Pool when there is no excess USDC due to high utilization rates. For example, two community members created a withdrawal mechanism via a [FIDU-USDC Curve pool ↗](#).*

Withdrawal Distributions

Withdrawal Requests are fulfilled every two weeks. If there is enough USDC unutilized in the Pool to honor all outstanding withdrawal requests at the end of the distribution period, all withdrawers will receive 100% of the value of the FIDU they requested to withdraw, in USDC.

If there is not enough USDC in the Senior Pool to honor all outstanding Withdrawal Requests at the end of the period, due to the USDC being actively utilized by Borrowers, all available USDC will be allocated to withdrawers pro-rata.

For example, if there is enough USDC in the Senior Pool at the end of the period to fulfill 80% of the current Withdrawal Requests, all withdrawers will receive 80% of the value of the FIDU they've requested to withdraw.

Withdraw Requests that are partially filled will be automatically rolled over into the next period, and one's Withdrawal Request remains active until it is completely fulfilled or canceled. An LP can only have one active Request at a time, although they can increase that request while it is active.

Once a Withdrawal Request has been made, it can only be increased or canceled, not reduced. If a Request is canceled before it is fully fulfilled, the LP is charged a cancellation fee of 1% of the total remaining request.

Withdrawal Request FAQs

Will I be able to participate in staking rewards while my FIDU is in the Withdrawal Request?

No, staked FIDU must be unstaked before it can be deposited to a Withdrawal Request.

Any staked FIDU that is supplied to a Membership Vault must be withdrawn from the Membership Vault before it can be un-staked. Then, it must be un-staked before it can be deposited to a Withdrawal Request.

As such, neither staking rewards nor Member Rewards can be earned while FIDU is in the Withdrawal Request.

Will I earn interest from the Senior Pool while I'm withdrawing?

Yes, FIDU that is in an active Withdrawal Request, including both new Requests and rollover Request balances, will continue to appreciate in value while it is pending withdrawal. LPs will still benefit from interest payments that the Senior Pool receives.

However, USDC that has been distributed to an LP at the end of a withdrawal period, but that the LP has not yet withdrawn from the dapp, will not accrue any further interest.

Why not a queue?

In a withdraw queue system, if a large FIDU holder submitted a large withdraw request they could reserve 100% of the incoming capital to the Senior Pool until their request was completely filled. This would prevent other FIDU holders from withdrawing any amount of their position.

How will this solution affect bots?

Speed of withdrawal will no longer be an advantage for withdrawing capital from the Senior Pool. Bots will need to submit a Withdrawal Request to withdraw, like any other LP.

Backer Position: PoolToken NFT

[Backers](#) with a [Unique Identity NFT \(UID\)](#) may withdraw interest payments as they are paid by the Borrower, which is monthly across all Goldfinch Borrower Pools as of August 2022. Principal liquidity is dependent on the Pool term as proposed by the Borrower, currently three years on average as of August 2022.

There is no withdrawal fee for claiming a Backer's share of a Pool's interest repayments or principal.

The Goldfinch dapp does not directly support PoolToken NFT secondary sales. However, community members and third-parties have launched various integrations via governance to provide Backers with the ability to sell their PoolToken NFTs. For example, [AlloyX launched a Goldfinch integration ↗](#), and the Goldfinch community is launching a [community-driven PoolToken NFT market ↗](#) that will allow backers to sell their PoolToken NFTs.*

**Please note that neither the FIDU-USDC Curve Pool, PoolToken NFT market, integrations, nor any other marketplace outside the Goldfinch dapp, are built and maintained by Goldfinch. Therefore, you should use your own judgment before selling FIDU or Backer PoolToken NFTs outside of the Goldfinch dapp.*

Default Process

What triggers a Borrower Pool to enter default status, and the process that follows.

All parameters for the default process are defined in the off-chain legal agreements established per Borrower Pool, in addition to being defined in the Pool's smart contract.

Borrowers provide performance reporting (e.g. covenant compliance, asset quality performance) directly to their Investors via the [Borrower Communication tooling ↗](#), which also allows Investors to engage directly with Borrowers throughout any default process that may take place.

1. Borrowers make monthly interest payments to their Pools to service their debt, as defined in the Pool terms.
2. When a Borrower misses an interest payment date, defined by their off-chain agreement, or breaks any other provision in their off-chain agreement:
 - **Off-chain:** Borrower enters a 3-7 (exact dates vary by pool according to what investors negotiated for) grace period to remediate the missed payment, or broken provision.
 - **On-chain:** Borrower enters a 45 day grace period before the protocol marks the loan on-chain as being in default, and starts provisioning the loan.
3. After 45 days of non-payment, the loan officially enters default status on-chain.
 - The Pool is now subject to its default interest rate as defined in the Pool terms:
 - For **Senior Pool investors**, the value of the loan will on a daily linear basis, be automatically written down over 120 days. If repayments are eventually made, then the writedown will be reversed accordingly.
 - For **Backers** in a given pool, no writedown mechanics are necessary since there is already a defined waterfall and pro rata distribution.
 - Investors work with the Borrower via Borrower Communication channels or off-chain methods to identify the conditions leading to the default, remediate and agree to a repayment plan, and establish transparent solutions to move forward with future repayments.
 - Any partial payments to the Pool are distributed first to the Senior Pool (senior tranche debt) and then to Backers (junior tranche debt).

4. If the remediation process is not successful, Goldfinch Investors will be able to pursue off-chain legal recourse via any on and off-chain overcollateralization the Borrower had provided in order to recoup losses.
 - Currently, all Pools on Goldfinch are fully collateralized with off-chain assets.

Governance

Governance is managed by a community DAO and has the ability to perform maintenance functions and parameter adjustments to the protocol via decentralized voting, including:

- Upgrading contracts
- Changing protocol configurations and parameters
- Selecting Unique Entity Check providers
- Setting the rewards and distribution of GFI
- Pausing protocol activity in the event of an emergency

Participation

Anyone can participate in governance by making proposals, discussing them, and participating in Snapshot votes.

Governance Forum

The Goldfinch community governance portal is located at <https://gov.goldfinch.finance/> ↗. This is a venue for the community to make and discuss proposals. More details on the proposal process are at the governance portal [here](#) ↗.

Snapshot Votes

Snapshot votes take place at <https://snapshot.org/#/goldfinch.eth> ↗. You can use your GFI balance to vote, and the results are calculated according to [Quadratic Voting](#) ↗. To ensure sybil resistance with quadratic voting, participants need a [Unique Identity NFT \(UID\)](#) in order to vote.

Governance Council

The Goldfinch Council has been set up to execute limited types of on-chain transactions based on the community Snapshot votes. The Goldfinch Council is a 6-of-10 multisig with 10 members who represent all stakeholders of the protocol:

- **Borrower** — [Gautam Ivatury ↗](#), Managing Partner at [Almavest ↗](#), one of the early Borrowers on Goldfinch. Almavest provides debt capital to high-performing companies in a variety of sectors globally.
- **Borrower** — [Alexandre Liege ↗](#), CEO of [Cauris ↗](#), one of the early Borrowers on Goldfinch. Cauris is a credit fund created to bring decentralized financing to fintechs in emerging markets.
- **LP/Backer** — [Deltatiger.eth ↗](#), an early Liquidity Provider and Backer on Goldfinch. Designed Synthetix monetary policy and active participant in many other protocols, including core team and multisig at [boot.finance ↗](#).
- **LP/Backer** — [Andrei Ansimov ↗](#), an early Liquidity Provider and Backer on Goldfinch. Also an early Flashbots contributor and active participant with many protocols.
- **Crypto Community** — [Manish Adhikari ↗](#), Goldfinch Discord Community Manager, and blockchain enthusiast.
- **Crypto Community** — [Viktor Bunin ↗](#), Protocol Specialist at Coinbase Cloud and active contributor to many crypto communities.
- **Crypto Community** — [Nat Robinson ↗](#), Co-Founder & CEO of [Leaf Global Fintech ↗](#), a company using crypto to provide affordable financial services for refugees in Rwanda, Uganda, and Kenya.
- **Early Team** — [Mike Sall ↗](#), Cofounder of [Warbler Labs ↗](#), which is part of the Goldfinch community and supports the growth and development of the Goldfinch ecosystem and broader DeFi space.
- **Early Team** — [Blake West ↗](#), Cofounder of [Warbler Labs ↗](#), which is part of the Goldfinch community and supports the growth and development of the Goldfinch ecosystem and broader DeFi space.
- **Early Team** — [Andrew Huelsenbeck ↗](#), Head of Operations at [Warbler Labs ↗](#), which is part of the Goldfinch community and supports the growth and development of the Goldfinch ecosystem and broader DeFi space.

Community-driven Proposal Tracker

Head on over [here ↗](#) to keep a track of current and upcoming proposals.

Tokenomics

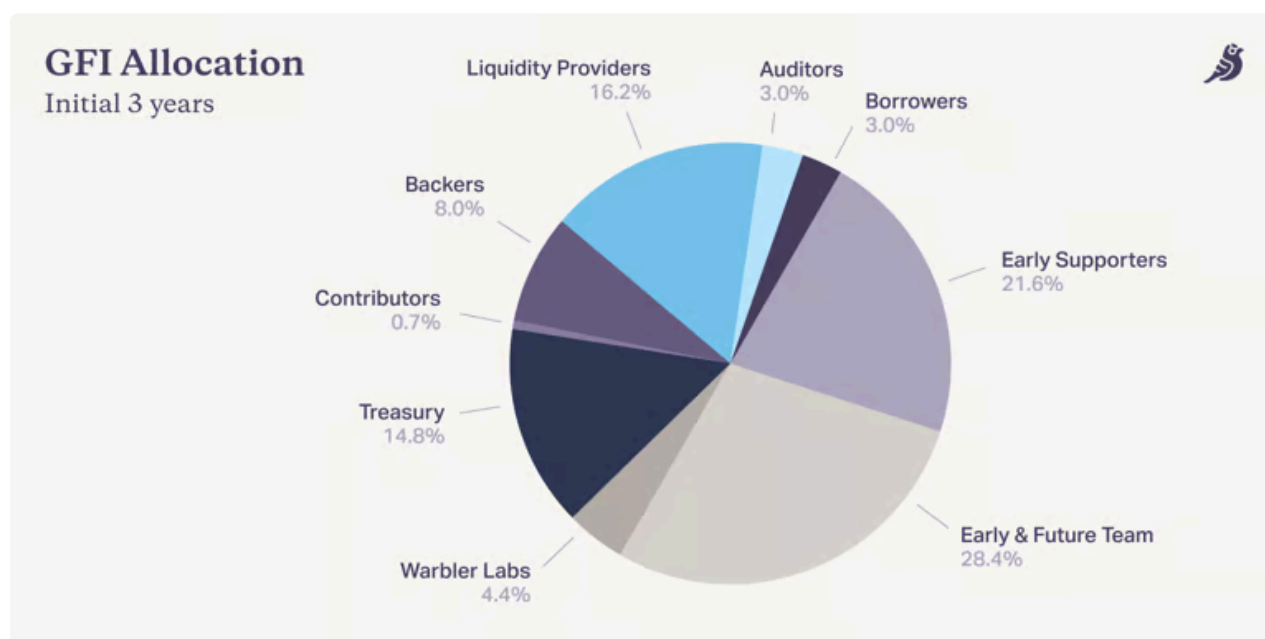
Total token supply

The initial token supply is capped at **114,285,714 GFI tokens**.

There is currently no inflation, but it is expected that it will be beneficial for the protocol to incorporate modest inflation after 3 years in order to reward future active participants. Ultimately, this will be up to the community to discuss and decide.

Token allocations

The initial allocation of the total supply of GFI are as follows:



Liquidity Providers (16.2%)

4.2% — Early Liquidity Provider Program: These tokens are allocated to the early Liquidity Provider program, which incentivized the very first participants to supply capital to the protocol. This program closed in July 2021. These allocations unlock over 6 months beginning on January 11, 2022, with a 12-month transfer restriction for U.S. participants.

4.0% — Retroactive Liquidity Provider Distribution: These tokens are allocated to all 5,157 liquidity providers as of a Dec 14 snapshot, excluding the Early Liquidity Provider program above. These distributions are only to non-U.S. persons and unlock over a range of immediate to 12 months, depending on the contribution amount and earliest contribution date.

8.0% — Senior Pool Liquidity Mining: These tokens are allocated to ongoing liquidity mining, beginning immediately. Senior Pool liquidity mining is described in the [Liquidity Mining](#) section.

Backers (8.0%)

3.0% — Flight Academy: These tokens are allocated to the 10,182 non-U.S. participants in Flight Academy. 2.85% are distributed [according to the tiers described in this post ↗](#), with an unlock schedule ranging from immediate to 24 months. The remaining 0.15% are allocated to future participants.

2.0% — Backer Pool Liquidity Mining: The Backer Pool liquidity mining system grants tokens to Backers as interest payments are made into Borrower pools, and the system is now in place, following a [governance proposal ↗](#). There have also been retroactive distributions for existing Backers who have already supplied to Borrower pools as well — [this proposal ↗](#) has more details.

3.0% — Backer Staking: These tokens are allocated for Backers who stake GFI on other backers, as described in the [whitepaper ↗](#). This is not yet live, but the community is expected to introduce and vote on a proposal for this in the coming months.

Auditors (3.0%)

3.0% of tokens are set aside for auditors, for any future auditor system launched by the protocol through decentralized community governance. An auditor system is not yet live, but we expect the community to introduce and vote on a proposal for one in the coming months.

Borrowers (3.0%)

3.0% of tokens are set aside for Borrowers, for when and if the community decides to implement a future distribution system for Borrowers.

Contributors (0.65%)

0.65% is allocated to contributors who have already significantly contributed to the community and protocol, either through a management role in the community Discord, by creating great art or memes, or through contracting agreements with the Foundation. Contributors who participated in Flight Academy will receive Flight Academy rewards as part of this category. These distributions generally follow the same unlock schedule as the Flight Academy distributions.

Community Treasury (14.8%)

14.8% is allocated to the community's treasury, which the community can decide to use for purposes such as grants to developers and contributors, adjustments to protocol distribution mechanics, and coverage for potential loan defaults.

Early and Future Team (28.4%)

28.4% is allocated to the early Goldfinch team of 25+ employees, advisors, and contractors. Full-time contributors are subject to 4 or 6 year unlock schedules, and part-time contributors are subject to 3-year unlock schedules, all with initial 6-month lock-ups and 12-month transfer restrictions.

Warbler Labs (4.4%)

4.4% is allocated to Warbler Labs, a separate organization spun out from the early Goldfinch team that will contribute to the Goldfinch community and broader DeFi ecosystem. The tokens are subject to a 3-year unlock schedule with an initial 6-month lock-up and 12-month transfer restriction.

Early Supporters (21.6%)

21.6% is allocated to a group of 60+ early supporters who invested \$37M to help build the protocol. These supporters are all long-term oriented and have 3-year unlock schedules, as well as an initial 6-month lock-up and 12-month transfer restriction.

Token Launch FAQ

How will I know if I received GFI?

Go to <https://app.goldfinch.finance/gfi> ↗. If you have received GFI tokens, you will see it listed with an button to accept them.

How can I receive more GFI?

You can receive more tokens through [liquidity mining](#) as a Liquidity Provider. Learn [how to participate in liquidity mining here](#).

How can I claim my GFI?

See [Claiming GFI distributions](#).

Why can't I claim all my GFI? What is 'unlocking'?

Some distributions are not all immediately available, but instead "unlock" according to a schedule. For example, a 2-year unlock schedule means you will gradually be able to claim more and more of your distribution, eventually being able to claim 100% after 2 years.

The frequency with which more of your GFI becomes claimable is the "unlock interval". For example, if the unlock schedule is monthly, that means new amounts of the GFI distribution will become claimable on a monthly cadence.

I was a Liquidity Provider in the Senior Pool before Token Launch. Will I get tokens?

Yes, if you are a non-U.S. person, and if you kept your capital in the Senior Pool until December 14, 2021 at UTC 10:00:17 PM, you will get GFI. Existing Liquidity Providers were allocated GFI distributions during the token launch — see how they were allocated on the [Tokenomics page](#). In addition, existing Liquidity Providers can receive more tokens through [liquidity mining](#) by staking their FIDU. Learn [how to participate in liquidity mining here](#).

I withdrew capital from the Senior Pool before the January 11, 2020 Token Launch. Will I get tokens?

All parties that had capital in the Senior Pool until December 14th, 2021 UTC 10:00:17 PM snapshot will receive GFI. The exact block: <https://etherscan.io/block/13805859> ↗. Withdrawal before this point means forfeiting of GFI rewards.

I was a community member on Discord. Will I receive GFI?

The only community members who received GFI during the Token Launch were those who held the "role names" of Community Managers, Admin, and Contributors for their excellence and support in handling the community. A few select Artists and Developers from the community also received GFI. However, in the near future, the community may want to set up community-run grants programs which will allow other community members to earn GFI.

Can GFI be staked?

No — GFI, the Goldfinch Protocol token, cannot be "staked" right now.

To avoid confusion, there are two unrelated types of "staking" on Goldfinch:

- When people supply capital to the Senior Pool, they receive [FIDU](#). They can then stake that FIDU to participate in Liquidity Mining and earn GFI.
- According to the [white paper](#) ↗, in the future GFI may be used for staking (on Backers), but this functionality is currently not in place.

Can I use GFI for anything while it's unlocking?

No — GFI can only be used once it is in your wallet. Sometimes, you may have GFI that is claimable, but not in your wallet. To transfer unlocked GFI to your wallet, [Claim](#) for the GFI on the [GFI](#) tab in the [dapp](#) ↗.

Can I transfer my GFI tokens to a Ledger?

Yes, after you have claimed your GFI. Ledger supports ERC20 tokens like GFI.

I was a Backer for one or more Borrower Pools. Will I get GFI?

The audit for the Backer GFI rewards was submitted in February 2022are now live, after proposals on [Backer Liquidity Mining ↗](#) was passed by the community. In addition, Backers will be able to participate in Staking Rewards liquidity mining, as per [another proposal ↗](#) passed by the community.

Unique Identity (UID)

Unique Identity (UID) is a non-transferrable NFT representing Know-Your-Customer (KYC), Know-Your-Business (KYB), and/or U.S. Investor Accreditation verification on-chain. It is the first NFT for identity. It follows ERC-1155 standards, and is freely usable by any other protocol. No personally identifiable data is stored on-chain.

Know-Your-Customer verification is a check that ensures a person is actually who they say they are, and Know-Your-Business verification is a check that ensures an entity and its representatives are actually who they say they are. Doing so is a legal requirement that prevents cases of fraud and any suspicious activities from occurring on the Goldfinch protocol.

U.S. investor accreditation is for U.S.-based individuals and entities who want to participate in the protocol. Under U.S. federal security laws, only persons who are accredited investors may participate in certain financial opportunities.

There are currently 5 types of UIDs:

1. ID_TYPE = 0, Non-U.S. Individual: can vote and supply capital across the protocol
2. ID_TYPE = 1, U.S. Accredited Individual: can vote and supply capital to the Senior Pool, and to Borrower Pools on a case-by-case basis (depending on Borrowers' requirements)
3. ID_TYPE = 2, U.S. Non-Accredited Individual: can vote, but cannot supply capital
4. ID_TYPE = 3, U.S. Accredited Entity: can vote and supply capital to the Senior Pool, and to Borrower Pools on a case-by-case basis (depending on Borrowers' requirements)
5. ID_TYPE = 4, Non-U.S. Entity: can vote and supply capital across the protocol

Why UID matters

Having an on-chain representation of personhood achieves two major unlocks. First, it opens DeFi to an entirely new set of real-world participants, notably companies and financial institutions. It also greatly expands the design space for new features and mechanisms on DeFi protocols.

Many companies today don't interact with DeFi due to KYC/KYB and counterparty concerns. But UID solves these issues for them. Many companies are already [borrowing through Goldfinch ↗](#), and they wouldn't be able to without the KYC/KYB and accreditation verifications. As companies grow comfortable with crypto, they could drive an incredible amount of new volume to DeFi protocols.

UID is completely interoperable with the rest of DeFi. It is a standard ERC-1155 contract, so any protocol can get the benefits without having to build their own KYC flow or handle any data themselves. If you're a builder, go to the [Developer docs](#) to learn more.

It's also important to emphasize that UID seriously factors in privacy. No personally identifiable data is stored on-chain. Goldfinch has partnered with [Persona ↗](#) to manage the KYC process and data, and with [Parallel Markets ↗](#) to manage the KYB and accreditation processes and data. They both offer [industry-leading privacy and security practices ↗](#) that are trusted by many of the largest technology companies. Learn more [here ↗](#).

How UID works

For Non-U.S. Individuals and U.S. Non-Accredited Individuals

When a user completes their KYC process, Persona checks that they have a valid identity and are not a duplicate of other ones. Once verified by Persona, the user is eligible to get his or her UID.

For Entities (Non-U.S. and U.S.)

When a user submits their KYB information, Parallel Markets checks that they have valid entity documentation and valid business owners. Once verified by Parallel Markets, an entity is eligible to get its UID.

For U.S. Accredited Investors

In addition to submitting KYC information to Parallel Markets, a user also submits information to prove his or her accreditation qualification(s). Parallel Markets checks that they have a valid identity, are not a duplicate of other identities, and that accreditation documents are valid. Once verified by Parallel Markets, the user is eligible to get his or her UID.

For All

Once eligible to get a UID, the user submits an Ethereum transaction that mints the UID, which is a non-transferrable NFT that is sent to their address.

There is a 0.00083 ETH fee (about \$3) in addition to gas, in order to cover the cost of Persona's service.

Then, when the user participates in protocols like Goldfinch, these protocols can check that they have this UID, ensuring they are unique and KYC'ed, KYB'ed, or accredited.

Get started minting a UID with the next guide, [For Users](#).

About the NFT Artist

The amazing illustration for UID was created by [Marcelo Meijome ↗](#). You can see some of his other [NFT works on Foundation ↗](#).

For Users

How to create your UID

As of January 26, 2022, U.S. individuals can obtain a Unique Identity (UID). This will allow US-based GFI holders to participate in [governance](#). However, only U.S. Accredited Investors will be able to participate in supplying capital to the protocol at this time.

Step 1: Sign In

Head over to the [account page](#) ↗. Click the “Connect Wallet” button and use MetaMask to sign in to Goldfinch with the MetaMask address you’d like to use to interact with and supply capital to the protocol.

Step 2: Verify your MetaMask address

Self-identify as an individual or entity, US-based or non-US, and—for US-based—accredited or non-accredited. For non-US individuals and US non-accredited individuals, you’ll then be prompted to enter your contact information and will be redirected to Persona—the Goldfinch protocol’s identity verification partner—to complete an identification check. Please have your ID ready and be prepared to take a quick selfie.

For entities (both US and non-US) and US accredited investors, you’ll then be prompted to visit the Goldfinch bridge site on Parallel Markets to complete the KYC/KYB process. Once your KYC/KYB (and accreditation, if applicable) are verified, you will get an email notification from Parallel Markets. If your identity and accreditation documents are accepted, you will be able to mint a UID on the [account page](#) ↗. If your documents are rejected, you can contact us at UID@warblerlabs.com for next steps.

Completing the minting step will allow you to participate in Goldfinch.

Step 3: Mint your UID

As mentioned above, you can go to the [account page](#) ↗ to mint your UID. To do this, click on [Create UID](#) and confirm the NFT minting transaction.

Note that minting a UID NFT will cost 0.00083 ETH + gas fees. This is to cover the fees incurred by the protocol for conducting an identity verification via Persona, a 3rd party service provider.

You can view your NFT after it's been minted by searching for your Metamask account on [Opensea](#) ↗. Your UID NFT will allow you to participate in Goldfinch as an LP and/or a Backer, and/or to participate in Governance as a GFI holder.

Step 4: Earn!

Head to [goldfinch.finance/earn](#) and select [Connect Wallet](#) to connect to the dapp using the same wallet you used to mint your UID. You can now invest in the protocol.

For Developers

This page provides technical guidance on how UID works, and how to integrate it with your protocol or dApp.

⚠ UID should be considered alpha software. It is used by the Goldfinch network, and we would love to see other protocols utilize the tool, but Warbler Labs cannot guarantee support or maintenance at this time. Use UID at your own risk. If you have questions or feedback, please contact us at uid@warblerlabs.com. We'd love to hear from you!

Integrating UID in your smart contracts

To integrate UID into your smart contracts, simply check the balance of your user's address to see if they possess the UID token. In Solidity, it would look like this:

```
# This is using OpenZeppelin's ERC1155 preset contracts:
#
https://docs.openzeppelin.com/contracts/3.x/api/token/erc1155#IERC1155
import "@openzeppelin/contracts/token/ERC1155/IERC1155.sol";

contract MyContract {
    address UID_CONTRACT =
"0xba0439088dc1e75f58e0a7c107627942c15cbb41";
    uint256 public constant ID_VERSION_0 = 0;

    function hasUID(address userAddress) returns (bool) {
        uint256 balance = IERC1155(UID_CONTRACT).balanceOf(userAddress,
ID_VERSION_0);
        return balance > 0
    }
}
```

Note: UID currently only operates on Ethereum mainnet. The contract itself can be found [here](#) ↗

Integrating UID in your dApp

For your front-ends, you can similarly integrate UID with the following

```
var Web3 = require('web3');
const web3 = new Web3(Web3.givenProvider);
const minimalAbi = [{"inputs": [{"internalType": "address", "name":
"account", "type": "address"}, {"internalType": "uint256", "name":
"id", "type": "uint256"}], "name": "balanceOf", "outputs":
[{"internalType": "uint256", "name": "", "type": "uint256"}],
"stateMutability": "view", "type": "function"}]
const uidAddress = "0xba0439088dc1e75F58e0A7C107627942C15cbb41"
const UID = new web3.eth.Contract(minimalAbi, uidAddress)

async function hasUID(userAddress) {
  const VERSION_0 = 0
  const balance = parseInt(await UID.methods.balanceOf(userAddress,
VERSION_0).call())
  return balance > 0
}
```

Getting new users verified

If your users don't already have a UID, you can send them to our front-ends to be verified at app.goldfinch.finance/verify.

How It Works

UID functions by having a trusted signer (currently Warbler Labs, the core development team for Goldfinch and creator of UID) verify that a given address has passed KYC via the third-party integration (currently Persona). Once verified, the user receives a signed message which they can present to the UID contract in order to mint their UID.

So the user flow looks like this...

1. User completes KYC flow (on app.goldfinch.finance/verify, via our partner [Persona](#)).
2. Submit completed KYC to the Trusted Signer, which returns a signed message to the user.
3. User mints their UID by presenting the signed message to the UID contract.

Security and Privacy

All personal data is processed and handled through [Persona](#) ↗. Persona has industry leading [security and privacy practices](#) ↗. **No personal information is stored on-chain.**

Need to get in touch?

Have questions about integrating? Do you have a certain use case, or feature you'd like to see? The community would love to hear from you. Reach out at **uid@warblerlabs.com**

Unique ID FAQ

For greater support, head over to the Discord server linked under the Important Links section.

What happens if I lose access to my UID? Can UID be reissued to a new wallet address?

Currently, it cannot. Burning and reissuing tokens is possible on the smart contract, though this has not been implemented on the front-ends. Theoretically, upon burn, the KYC/KYB/accreditation information would need to be deleted and then members could sign up with a new wallet.

Do I need to undergo verification for deposit and withdrawal in both the Senior Pool as well as Borrower Pools?

Yes, this is mandatory. UID is the first NFT for identity. No personally identifiable data is stored on-chain. More on UID [here](#).

How can I verify my identity?

Before minting UID NFT, you need to verify your Ethereum address from here: <https://app.goldfinch.finance/verify>. Please head over to the Tutorials section for a detailed explanation.

How do I know whether I am verified?

You can go back to <https://app.goldfinch.finance/verify> and sign in. If you are verified, you will see the message - “Your address verification is complete” or “Your verification was approved to immediately access the Senior Pool”.

Why do the Backers need to verify their identity/be KYC'd?

The protocol must have sybil resistance in order to implement the "trust through consensus" mechanism outlined in the whitepaper. Technically, this does not need to be KYC, and the community hopes to offer versions of this in the future which do not require ID verification. However, KYC offers sybil resistance as well as the legal compliance which many of the Borrowers require for their own purposes.

Note, Goldfinch's KYC/KYB and accreditation is handled by third parties, Persona and [Parallel Markets ↗](#), who, like Goldfinch, take privacy and security [very seriously ↗](#).

Is UID interoperable with other protocols?

UID is completely interoperable. It is a standard ERC-1155 contract, so any protocol can get the benefits without having to build their own KYC flow or handle any data themselves. If you are a builder, check out the docs here to learn how to integrate it.

Can I transfer my UID to others or can I sell my UID on OpenSea?

Because the UID is an ERC-1155 contract, it is not transferable and cannot be sold.

Borrower Communication

Tooling to facilitate communications between Borrowers and Investors on Goldfinch

In response to demand for more efficient channels of communication between Investors and the Borrowers on the protocol, the community introduced standardized channels to allow Borrowers to share updates on their pools and to communicate directly with the protocol's Investors.

This new Borrower Communications tooling is the first of its kind to use web3 solutions for investor reporting, and provides even more transparency for Goldfinch Investors by providing them with easy access to information like:

- Trusted, secure, real-time communication channels between Borrowers and Investors
- Accessible way for Borrowers to fulfill any and all obligatory Investor Reporting requirements, creating easier inroads to access the reporting that exists in their off-chain documents and agreements
- More effective channels for Investors to securely communicate with one another, enabling enhanced collective bargaining power.

For each Borrower Pool on Goldfinch, there will be two channels launched:

1. A **token-gated Discord channel** where the Borrower can communicate with Investors (both Backers and Senior Pool LPs) to provide updates, and where Investors can discuss the Pool amongst themselves. This will be managed by the Goldfinch community.
2. A **password-gated Dataroom** where Borrowers will post all necessary documents that they are obliged to provide as part of their off-chain legal agreements with Investors for compliance, reporting, and evaluation. This will be managed by the Borrower in line with what their existing real-world agreements require.

Each channel has a unique configuration to ensure that Investors are able to easily access the differing levels of information they need from Borrowers, while ensuring the confidentiality of the Borrowers' proprietary information is accommodated.

Channel #1: token-gated Discord channel

For every Borrower Pool on the Goldfinch protocol, there will be a **community-managed** Discord channel launched in the [Goldfinch Discord](#) ↗. This channel will be configured to permit access for:

- Backers invested in the Borrower Pool in question, who possess at least one pool token representing the Backer's participation in the Borrower Pool
- Liquidity Providers in the Senior Pool who possess at least 10,000 FIDU tokens or staked FIDU tokens, representing approximately a \$10,000 investment to ensure that the LP's accessing this proprietary data have adequately aligned incentives to protect a Borrower's best interests.

 **Note:** The rationale for a \$10,000 minimum to participate as a Senior Pool LP is driven by the need to ensure Senior Pool Investors meet an adequately high bar for "skin in the game." The incredibly sensitive nature of the underlying data Borrowers are reporting can essentially be thought of as trade secrets, given the depth of detail and private company data the Borrowers are reporting in these channels. With Borrowers forming a core component of the Goldfinch protocol, it's imperative that their interests are protected so that bad actors cannot easily the Borrowers' sensitive data.

The token-gating configuration is executed using the [guild.xyz](#) ↗ tool to grant access only to Backers and Senior Pool LPs as described above.

The primary purpose of this channel is to facilitate asynchronous communication between Borrowers and Investors. These communications can take a number of forms, including:

- Borrowers informing the capital providers that an update has been made to the Pool's Dataroom
- Borrowers answering questions posted by the Investors in the channel
- Borrowers confirming to Investors that an interest payment has been made

Channel #2: password-gated Dataroom

Each Borrower Pool will also launch a password-gated Dataroom where Borrowers can share documents pertinent to the Borrower Pool in question. This will be managed by the **Borrower**.

At a minimum, the Dataroom for each pool will contain the following documents:

1. All fully executed legal documents related to the specific pool
2. All waivers, amendments, and default notices
3. [Covenant](#) ↗ compliance certificates
4. Quarterly investor updates
5. Portfolio level reporting

It is worth noting that the community has defined minimum standards here in place of uniform reporting requirements as the wide and growing variety of borrower types on the protocol would make it difficult to enforce any stringent documentation requirements. In the future as new pools open, Investors can require specific documentation that might not fit every Borrower be provided as needed.

The access instructions for the Dataroom will be contained as a pinned message in the token-gated Discord channel for the pool in question. That way, any Backer or LP who accesses the Discord channel using their pool token or FIDU as verification will be able to easily access the Dataroom password as well.

✓ For step-by-step instructions on how to access these channels, view "[Accessing Borrower communication channels](#)" in the [Investor How-To](#) section of the documentation.

Investor How-To

Educational guides on how to use the protocol, for Goldfinch Investors

Getting Started on Goldfinch

The Goldfinch decentralized application (dapp), app.goldfinch.finance ↗, is how Investors, Borrowers, and the rest of the Goldfinch community participate in the protocol.

There are only a few high-level steps required to interact with the dapp to start investing:

1. **Navigate to** app.goldfinch.finance ↗
2. **Click "Connect Wallet" on the top right corner:** Currently, the Goldfinch dapp supports WalletConnect or Metamask. If you do not have a wallet, click "I don't have a wallet" to set one up.
3. **Verify your identity:** We'll cover how to mint a UID in the next page, [Verifying Your Identity](#).
4. **Explore the dapp:**
 - a. View all of the Borrower Pools and a summary of your holdings on the [Earn page](#) ↗
 - b. View the Senior Pool page, and [participate as a Liquidity Provider](#)
 - c. View the Borrower Pool pages, and [participate as a Backer](#)
5. **(Highly encouraged) Join the [Discord](#) ↗!** The Discord is a great place to ask the community for help with any questions you have along the way, stay in touch with announcements about different [Borrower Pools](#), and generally participate in the community.

Ready to get started? Move on to Verifying Your Identity on the next page.

 **Goldfinch provides tools to help participants interact with the protocol, but it is your responsibility to stay compliant with all legal, regulatory, and tax laws for your relevant jurisdiction(s). These guides are not financial advice, but are instead educational guides for how to use the Goldfinch decentralized application. All protocol participants are strongly encouraged to consult with their own legal and tax counsel.**

Verifying Your Identity

It is necessary to verify your identity in order to participate in Governance votes or supply capital to Goldfinch.

U.S.-based individuals must also verify their Accredited Investor status in order to supply capital.

The [identity verification process](#) checks that you are a real individual, and then allows you to mint your [UID](#). The process uses [Persona](#) ↗ and [Parallel Markets](#) ↗ to perform the third-party verification, which takes privacy and security [very seriously](#) ↗.

Step 1: Verify your address

1. Go to `https://app.goldfinch.finance/verify`. If you visit the Senior Pool or Borrower Pool pages before verifying, they will prompt you to first verify on this page.
2. In the `Verify your address` section, identify whether you are an individual or entity, US or non-US based, and—for US-based—whether you are an Accredited Investor or not.
3. **For Non-U.S. Individuals and U.S. Non-Accredited Individuals:** you will be prompted with Verify ID to start the process with Persona. For individuals, you will be asked to upload a passport or national ID documents. **Passports are strongly recommended as Persona currently does not support some forms of national IDs.**
4. With Persona: After uploading your ID, you will see the `Let's make sure it's you` pop-up and you will be prompted to take photos of the front and sides of your face.
5. Done! The page should say you have successfully verified your address.

 Disclaimer: Generally, the address verification process is instant. However, sometimes it can take some time for individuals to get verified. To ensure that you are instantly verified, see the "Helpful Tips" section below.

For Entities (both U.S. and Non-U.S.) and U.S. Accredited Individuals:

You will be prompted to visit the Goldfinch bridge site with Parallel Markets.

For individuals, you will be asked to upload a passport or national ID documents.

Passports are strongly recommended as Persona currently does not support some forms of national IDs. Entities will have to supply additional business documentation and information.

With Parallel Markets:

After submitting your information, Parallel Markets will typically process your information within 72 hours.

-  Once your KYB/KYC and Accreditation (if applicable) are verified, someone from the Goldfinch community will reach out within 24 business hours to collect the wallet address that your UID will be associated with. Once the address is provided, you will receive a second notification with confirmation that you can proceed with minting your UID.

Step 2: Mint your UID

Learn more about UID [here](#).

Please note you will need enough ETH in your address to cover 0.00083 ETH + gas fees. The additional 0.00083 ETH helps cover the fees for the Persona service.

1. Once you have completed your address verification (described above), you will be able to mint your UID.
2. Navigate to app.goldfinch.finance and select **Connect Wallet** to connect the same wallet to the dapp that you associated with your UID in Step 1.
3. On <https://app.goldfinch.finance/verify>, click **Create UID**.
4. Click **Confirm** in the popup that appears to verify the transaction in your wallet.

Advanced: You can check the gas fee and adjust it so that the transaction is as affordable for you as possible. [This tracker](#) ↗ can help you find low gas fees. [Click here](#) ↗ to learn how to adjust it.

5. Tada! Your UID will get minted once the transaction is confirmed. You can view yours by connecting your wallet to [Opeansea](#) ↗.

Helpful tips for Persona

Your identity is tied to your Ethereum account address, and you can only have one verified account. Unfortunately, there are some regions, like China, that the Goldfinch protocol currently does not support.

Here are some tips to ensure that your verification is successful on the first attempt.

- Ensure that the Identification Document (ID) you are using is an ID type that is accepted by Persona.
- Passport is strongly recommended as an identification document, since Persona may not recognize some local IDs.
- Make sure your ID has a full legal name (given name + surname), a recent picture, your birthdate, and is not expired.
- When using a residence permit as an ID, make sure you take pictures of both front and back.
- Use a phone or computer to take a picture of your ID and do not run it through an image editing software.
- If you have registered on Persona once using a different Ethereum address, you will not be able to register again.
- Some older forms of IDs, like domestic passports, may not work.

If you still run into issues, please message in the "#verification-support" channel on the community-run [Discord](#) ↗.

Now that you have your UID and are connected to the dapp, read on through the rest of the guides to participate in investing on the protocol.

Participating in Callable Deals

Callable loans provide Backers with the flexibility to call back their capital with 2 months notice.

Backers evaluate Callable Deals and provide capital to a single tranche of the deal itself. This new deal went live in March 2023, after community and governance approved the new pool type.

Learning about a Callable Deal

How Callable Deals work

Callable loans are an existing structure in traditional finance that provides liquidity to investors by giving them the right to “call back” their capital at regular intervals.

On Goldfinch, the Callable Deal loan structure gives Backers the right to “call back” their invested capital. Borrowers are required to return 100% of this “called capital” at the end of a “call period.”

For the initial design, call periods will occur every three months and call requests must be submitted **at minimum 2 months** before the end of the call period (any calls that occur less than 2 months before the closest upcoming repayment date will be paid on the second closest upcoming repayment date).

Example Callable Deal

Callable loans are a suitable option for borrowers with short-term assets that can be liquidated when investors want to redeem their capital. Callable loans are used in traditional finance, but their structure and terms are usually customized to match the capital structure of the Borrower.

As an example deal:

- Borrower: Fintech X, who provides buy-now-pay-later (BNPL) loans to customers with a maximum tenor of 30 days

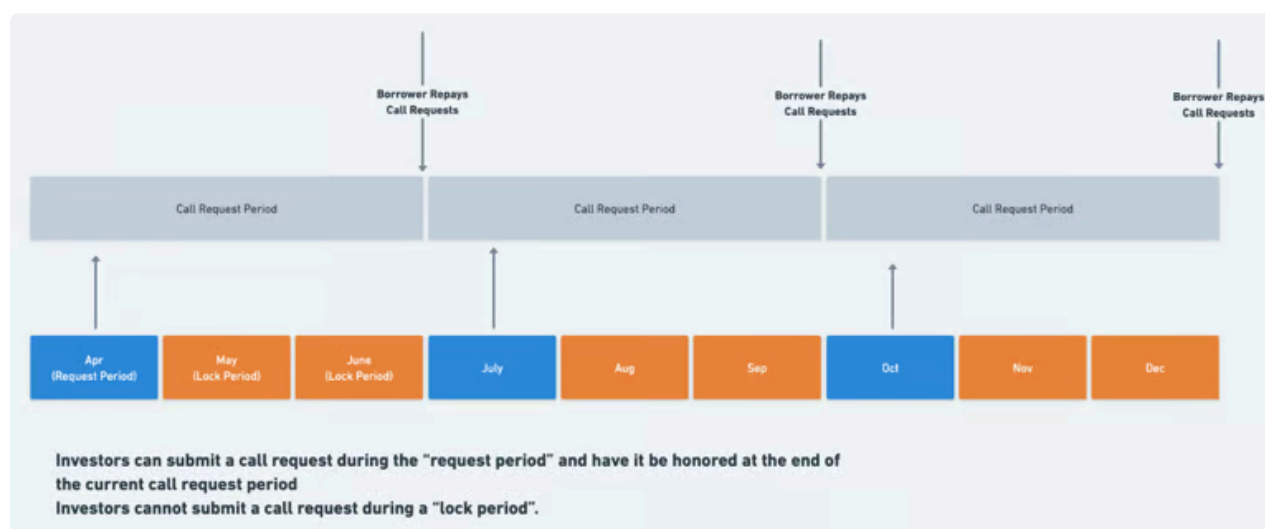
- Goldfinch Deal Terms: 2 year loan with a quarterly call period where Backers must serve their call notice at least 2 months in advance

Fintech X borrows from Goldfinch in order to offer more BNPL loans. If a Backer calls back their position early, Fintech X will know at least 2 months in advance. So they wait to get repaid on their 30 day BNPL loans and use that cash to repay Backers' called capital.

Recalling Principal

As mentioned above, Backers with UID can submit call requests to recall any portion of their principal at any point in time — but must be done at least 2 months in advance to call the next repayment. Once a request is submitted, the Backer continue to receive interest on capital until it is withdrawn.

The illustration below shows a sample deal that demonstrates how the payback period works.



Call submission scheduling

For example: If you submit a call on the earliest possible date in the window (May 1), the time-to-repayment is five months; if you submit on the last possible day of the period (July 31), the time-to-repayment is two months. If a borrower is late on any payments, the ability to submit call requests will remain active.

Understanding the risks

It is important to understand that you can lose money by participating in Callable Deals.

If a Borrower doesn't repay the amount they borrow, you will lose your relative portion of that amount. The Borrower repayments are applied pro-rata to outstanding balances in the following order:

1. All existing interest obligations
2. All existing call request principal obligations **which have been locked in**, in order of the call request period they were submitted in.
3. All accrued interest up until now
4. Remaining principal balance

Notes: Since initial Callable loans will be bullet loans, in most cases only (1), (2) & (3) apply; in the case of early principal repayments, the full waterfall applies, including (4). The addition of (2) in the repayment waterfall differs from our current deals.

When Callable Deals are open

You can only participate in a Callable Deal when the Callable Deal is open. Otherwise, the dapp will show that the pool is Full and you will no longer be able to supply capital. You can look out for announcements to see when new Borrower Pools are going to be open by <https://deals.goldfinch.finance/> ↗.

Announcements

Typically, once a Callable Deal is open, the community shares announcements using [Discord](#) ↗, Twitter, and an email list. Once the deal is live on the dapp, the deal page typically holds the key terms of the loan agreement, a credit memo, a data room, the repayment schedule, FAQs, and other resources.

Getting more details about Callable Deals

Usually, the deal page on the dapp includes both a credit memo and a data room with Non-Disclosure Agreements for you to sign. (Read more about NDA's [here](#) ↗.) This will give you access to the credit memo, produced by third-party professional credit analysts, and the due diligence data room containing information provided by the Borrower.

Each individual Goldfinch Backer is responsible for doing thorough due diligence on their own investments and deciding whether it's a good opportunity for them.

In addition, after you sign the NDA, there is usually a deal-specific Telegram group and/or Discord channel you can join. There, you can ask Borrowers questions directly, both about who they are and the specifics of the deal.

Supplying to a Callable Deal

Each Callable Deal has a unique set of terms and requirements, but all will require a UID to access ([Verify your identity](#) ↗ to mint your UID).

1. When the pool is open, visit <https://app.goldfinch.finance/earn> ↗ and click on Callable Deal under the list. Note: when a Callable Deal is full and closed, it will move to the section.
2. On the deal page, click to view the form. Review the agreement that is linked to understand what you are signing, enter your full name (this is your signature), and enter the amount of USDC you want to supply. Note: You will be required to enter your full legal name as was provided when registering your UID, and this name will be added to the Borrower's off-chain transaction agreement. This gives you all the associated rights, benefits, and security of the transaction you are investing in. Sometimes, pools will have a cap on the maximum amount that can be supplied. You may need to adjust your investment amount accordingly.
3. Review the terms and click .
4. Approve the transaction in your wallet.
5. Done. You should now see the USDC balance you supplied in your balance at the top of the page. Note: When you supply to a Borrower Pool, you receive an NFT that acts a "digital receipt" representing the amount you supplied. Learn more about this in the [Backer](#) ↗ section of the documentation.

Call Requests

After you supply capital, you can withdraw up until the time when the deal is closed (which typically happens once when the deal is full).

After the Borrower draws down the capital, Backers can submit call requests to recall their principal at any point of time. Repayment of call requests happens once every three months on a set date.

Timing

Call requests must be received in the first month of the three month call request period (two months prior to a repayment date) in order to be paid. Any calls that occur less than two months before the closest upcoming repayment date will be paid on the second closest upcoming repayment date.\

How To Submit a Request

1. Navigate to the deal's page in the dapp.
2. Click `Submit Call Request` and enter the amount of USDC you want to call back from the deal. Note the expected repayment date.
3. Click `Submit`.
4. `Accept` the transaction in your wallet.

How to Withdraw Capital

1. Return to the deal's page in the dapp on the expected repayment date.
2. Click `Withdraw`
3. Click `Submit`
4. `Accept` the transaction in your wallet.
5. Done. Following network transaction times you should see the USDC you withdrew in your wallet.

Interest bearing balance

Interest is generated by the product of the interest APR, time elapsed, and the amount of "interest bearing balance".

Interest bearing balance begins as the total amount of principal borrowed by the borrower. Afterwards, interest bearing balance decreases by the amount of repaid principal. However, if the borrower makes an early principal repayment, *interest bearing balance is only decreased after repaid principal would become due.*

As an example, let's say a borrower borrows \$1,000,000 from a Callable Loan in January. In March, if the borrower makes an early principal repayment of \$250,000, they still owe interest on the full \$1,000,000 amount (the "interest bearing balance") until the next principal due date. So in this case, they would owe interest on \$1,000,000 until May 1st. Starting May 1st, they would only owe interest on the new \$750,000 amount (the new "interest bearing balance").

**** Why? **** In order to make sure there is no incentive to game the timing of call requests, we only update owed principal to lenders at the end of a call request period. In order to ensure all borrowers are paid interest pro rata regardless of the time of their call submission within a call request period, we only update the interest bearing balance at the end of a call request period. Otherwise, earlier call requesters would effectively get less interest even though they were not allowed to withdraw earlier.

FAQ

What cryptoassets can I supply to Callable Deals?

Callable deals only support USDC investment.

When do I receive yield?

You receive yield each time the Borrower makes an interest payment into the Borrower Pools.

When do I receive principal?

You can call any portion of their principal at any point in time — but must be done at least 2 months in advance to call from the next repayment. Once a request is submitted, you will continue to receive interest on capital until it is withdrawn.

Do I need to sign a Non-Disclosure Agreement (NDA) to invest as a Backer?

Technically no, but it's strongly recommended that you do sign the NDA in order to review the credit memo and full due diligence materials about the Callable Deals you participate in.

What happens if I provide a fake name when I supply capital?

You will forfeit your legal rights to the real-world loan document you are signing. You will still be able to supply capital into the Callable Deals and receive your portion of repayments made on-chain. However, you will have no real-world legal recourse. The Borrower and other Backers will additionally have no obligations to make any repayments to you in the event that there is a default and they come to a decision regarding how to resolve it.

Are Backers able to take legal action if Borrowers default?

Yes, if there is a real-world legal agreement between the Borrower and Backers. When you enter your full name, you are directly signing the agreement with the Borrower that is linked in the form. The process by which a Backer can take action against a Borrower will depend on the specific real-world documents related to the pool a Backer invested in.

Participating in the Senior Pool

This section covers instructions on supplying / staking in the Senior Pool

[Liquidity Providers](#) participate in the the Senior Pool to receive yields with ease from automatic capital allocation across the protocol. The Senior Pool is a smart contract that automatically allocates capital across all Borrower Pools according to the consensus those pools receive from Backers.

Learning about the Senior Pool

How the Senior Pool works

Read about the Senior Pool on the [Liquidity Providers page](#).

Understanding the risks

It is important to understand that you can lose money by participating in the Senior Pool. If a Borrower never pays back the amount they borrow, you will lose your relative portion of that loan default. However, if a Borrower pays back a portion of what they owe, that payment will go to the Senior Pool first, before it starts to cover any losses for Backers. In this way the Senior Pool provides more protection, compared to participating directly in Borrower Pools as a Backer.

Understanding the APY

The APY shared at the top of the [Senior Pool page](#) ↗ is based on a current snapshot in time, and changes depending on two factors:

1. The Senior Pool's current usage. All interest payments from Borrower Pools is automatically allocated the Senior Pool. As more capital is supplied to the Senior Pool, the effective yield will decline as the repayments are spread across more capital.
2. How much capital the Senior Pool has deployed across Borrower Pools. As more capital is deployed, increasing utilization, there will be more repayments, thereby increasing the Senior Pool's APY.

The APY presented is a dynamic number that will change both as Liquidity Providers supply/withdraw from the pool, and as Borrowers make drawdowns and repayment from the Pools.

Supplying to the Senior Pool

1. Go to <https://app.goldfinch.finance/pools/senior>.
2. Ensure your wallet is connected by clicking .
3. Ensure you have [verified your identity](#).
4. Click and enter the amount of USDC you would like to supply to the Senior Pool.
5. Review the terms and check off ☐ I agree to the [Senior Pool Agreement](#).
6. If you check off ☐ I want to stake my supply to earn GFI rewards (some additional APY), you will receive additional GFI distributions. You can also choose to stake in a separate transaction after you supply to the Senior Pool, by following the [Participating in Liquidity Mining](#) guide.
 - Please note: Staking does not 'lock' your USDC; you can always withdraw at any time. Staking allows you to receive GFI distributions, which unlock over the first 12 months after staking. For more details, see [Senior Pool Liquidity Mining](#).
 - You can see your GFI rewards received for staking FIDU in the tab.
7. Click .
8. Accept the transaction on your wallet.
9. Done. You should now see the additional USDC in your balance at the top of the page.

Withdrawing from the Senior Pool

 Note: There is a 0.5% fee when withdrawing from the Senior Pool

1. Go to <https://app.goldfinch.finance/pools/senior>.
2. Click and enter the amount of USDC you want to withdraw
 - a. Please note: If you previously staked your FIDU, withdrawing will also unstake your FIDU.

- b. You will receive this amount minus a 0.5% fee that goes towards protocol reserves. The dapp will state what amount you will receive in the UI below the amount you stated.
3. Click `Submit.`
4. `Accept` the transaction in your wallet.
5. Done. Following the network transaction times you should see your USDC in your wallet.

Viewing your FIDU in Metamask

1. FIDU should appear in the assets tab on Metamask
2. If it doesn't appear, click import token at the bottom of the Assets tab, then custom token. Input the FIDU token address, which is `0x6a445e9f40e0b97c92d0b8a3366cef1d67f700bf`, and click `Add Custom Token`. It should now appear in your assets list.

FAQ

Can I withdraw immediately from the Senior Pool?

Yes, as long as there is enough USDC remaining in the Senior Pool. If the Senior Pool is 100% utilized and there is no USDC remaining, you will need to wait for other Liquidity Providers to supply first, or for Borrowers to make repayments.

What cryptoassets can I supply to the Senior Pool?

The Senior Pool only supports USDC.

When do I receive yield?

You receive yield each time Borrowers make interest payments into their Borrower Pools. Borrowers are typically on monthly payment schedules, and different Borrowers make their payments at different times in the month.

What is FIDU?

FIDU is like a "digital receipt" that represents how much you have supplied to the pool. It is very similar to [Compound's cTokens ↗](#). For more information, see [FIDU in Liquidity Providers](#).

Participating in Borrower Pools

[Backers](#) evaluate Borrower Pools and provide first-loss (junior tranche) capital to the Pools.

Learning about Borrower Pools

How Borrower Pools work

Read about investing in Borrower Pools on the [Backers page](#). Read about Borrowers and their incentives on the [Borrowers](#) page.

Understanding the risks

It is important to understand that you can lose money by participating in the Borrower Pools. If a Borrower doesn't repay the amount they borrow, you will lose your relative portion of that amount. By participating as a Backer, you are taking higher risk by providing first-loss capital—when a Borrower pays back a portion of what they owe, that payment is first applied to the Senior Pool before it is applied to Backers. Therefore, it is possible for Borrowers to make partial repayments and for you to still lose 100% of the amount you supplied.

When Borrower Pools are open

You can only participate in a Borrower Pool when a Borrower Pool is open. A pool can be open either because 1) a new Borrower pool is raising capital, or 2) a portion of an existing Borrower Pool has opened up. Otherwise, the dapp will show that the pool is Full and you will no longer be able to supply more capital. You can look out for announcements to see when new Borrower Pools are going to be open by [subscribing to the Backer Updates newsletter ↗](#).

Announcements

Typically, once a Borrower Pool is open, the community shares announcements using [Discord ↗](#) and/or Telegram. This announcement will contain a [Pool Overview ↗](#) (an example is linked), which describes the Pool deal in depth. It typically holds the key terms of the loan agreement, the repayment schedule, FAQs, and other resources.

Accessing Credit Analysis Memos

The Credit Analysis Grant is a community-led grants program, [approved by the community in July 2022 ↗](#), to compensate skilled community members to write Credit Analysis Memos for borrower pools. The credit memos do not contain any investment advice. For more details, please [visit this FAQ page ↗](#).

Each credit memo will be an NDA-gated, multi-page report that follows the same specific format: 1) summary, 2) overview, and 3) credit analysis. The credit analysis section will include details on historical financials, portfolio performance, deal structure, and risk.

The Credit Analysis Memos will be displayed on the dapp within their respective Borrower Pool pages. Users will have to sign into an NDA in order to access the report. Please note, as of February 6, Credit Analysis Memos are only available for the [LendEast ↗](#) and the [Cauris #2 ↗](#) Borrower Pools.

Accessing the Data Rooms for more details

Usually, the Pool Overview includes a Non-Disclosure Agreement for you to sign. (Read more about NDA's [here ↗](#).) This will give you access to the due diligence data room containing information provided by the Borrower. Each individual Goldfinch Backer is responsible for doing thorough due diligence on their own investments and deciding whether it's a good opportunity for them.

In addition, after you sign the NDA, there is usually a deal-specific Telegram group and/or Discord channel you will be invited to join. There, you can ask Borrowers questions directly, both about who they are and the specifics of the deal.

Supplying to Borrower Pools

Each Borrower Pool has a unique set of requirements, which are announced on the Goldfinch Discord server. But typically, the steps are:

1. Join [Discord](#) or [Telegram](#) and keep an eye out for announcements about Borrower Pools opening. If you aren't sure whether there were recent announcements, ask in the Discord.
2. [Verify your identity](#).
3. When the pool is open, visit <https://app.goldfinch.finance/earn> and click on the pool in the `Borrower Pools` list.
 - a. Note: when a Borrower pool is full, it will say "full" in the top right corner of the row.
4. On the pool page, click `Supply` to view the form. Review the agreement that is linked to understand what you are signing, enter your full name (this is your signature), and enter the amount of USDC you want to supply.
 - a. Note: You will be required to enter your full legal name as was provided when registering your UID, and this name will be added to the Borrower's off-chain transaction agreement. This gives you all the associated rights, benefits, and security of the transaction you are investing in.
 - b. Note: Sometimes, pools will have a cap on the maximum amount that can be supplied. You may need to adjust your investment amount accordingly.
5. Review the terms and click `I Agree`.
6. Approve the transaction in your wallet.
7. Done. You should now see the USDC balance you supplied in your balance at the top of the page.
 - a. Note: When you supply to a Borrower Pool, you receive an NFT that acts a "digital receipt" representing the amount you supplied. Learn more about this in the [Backer](#) section of the documentation.

Withdrawing from Borrower Pools

 After you supply capital, you can withdraw up until the time when the Borrower Pool is closed (which typically happens once when the Pool is full). After the Borrower draws down the capital, you will not be able to withdraw until the Borrower makes repayments. As the Borrower makes repayments, you will be able to withdraw your relative portion of those repayments.

1. Navigate to the Pool's page in the dapp.
2. Click **Withdraw** and enter the amount of USDC you want to withdraw from the Pool.
3. Click **Submit**.
4. **Accept** the transaction in your wallet.
5. Done. Following network transaction times you should see the USDC you withdrew in your wallet.

FAQ

What cryptoassets can I supply to Borrower Pools?

Borrower Pools only support USDC investment.

When do I receive yield?

You receive yield each time the Borrower makes an interest payment into the Borrower Pools.

Do I need to sign a Non-Disclosure Agreement (NDA) to invest as a Backer?

Technically no, but it's strongly recommended that you do sign the NDA in order to review the full due diligence materials about the Borrower Pools you participate in.

What happens if I provide a fake name when I supply capital?

You will forfeit your legal rights to the real-world loan document you are signing. You will still be able to supply capital into the Borrower Pool and receive your portion of repayments made on-chain. However, you will have no real-world legal recourse. The Borrower and other Backers will additionally have no obligations to make any repayments to you in the event that there is a default and they come to a decision regarding how to resolve it.

Are Backers able to take legal action if Borrowers default?

Yes, if there is a real-world legal agreement between the Borrower and Backers. When you enter your full name, you are directly signing the agreement with the Borrower that is linked in the form. The process by which a Backer can take action against a Borrower will depend on the specific real-world documents related to the pool a Backer invested in.

Is it necessary for the Backer portion of the pool to be filled before the Borrower locks the pool and draws down capital?

No, the Borrower can lock the pool and drawdown at any time. Moreover, in theory, it is also possible for the Backers to fill beyond the limit, which would reduce the leverage and expected APY for Backers. The limit displayed on the dapp is just imposed on the frontend as a way to ensure the APY stays at the expected level.

Participating in Liquidity Mining

Senior Pool Liquidity Mining

Currently, liquidity mining is only live for the Senior Pool. Learn more about how [Senior Pool Liquidity Mining works here](#).

If you're participating in the Senior Pool for the first time:

1. See how to [participate in the Senior Pool here](#).
2. In the Supply form, make sure to check the box to stake your position within the same supply transaction.
3. Once you complete the supply transaction, you will start receiving GFI distributions from Senior Pool liquidity mining.

If you have already supplied to the Senior Pool and have FIDU:

1. On the [Senior Pool page ↗](#), you should see a prompt describe the unstaked FIDU you have with an option to "Stake All FIDU".
2. Click on the "Stake All FIDU" button to approve and stake your unstaked FIDU.
3. Once you complete the Stake transaction, you will start receiving GFI distributions from Senior Pool liquidity mining.

View your GFI

You can view the GFI you receive from liquidity mining on the GFI tab. Learn more about [claiming GFI here](#).

Claiming GFI Distributions

This is a manual for receiving GFI.

Claiming GFI

1. Go to <https://app.goldfinch.finance/gfi>.
2. This page will list all of the GFI you have received
3. If a particular GFI distribution has an unlock schedule, you will need to first **Accept** it. This will open a transaction to accept in your wallet.
 - a. Note: By accepting, you will mint an NFT that automatically manages the unlock schedule
 - b. After accepting, it should now show a **Claim** button. You will be able to claim more GFI over time as it unlocks.
4. Click **Claim** to claim the GFI. This will open a prompt, where you can click **Claim** again.
5. Accept the transaction on your wallet.
6. Done. You should now see the GFI in you wallet balance.

The difference between "Accept" and "Claim"

Accept: When a GFI distribution has an unlock schedule, you need to "Accept" that distribution before being able to claim it. By accepting, you will mint an NFT that automatically manages the unlock schedule. After accepting, you will be able to claim the GFI as it unlocks.

Claim: When GFI is immediately available (and unlocked, if it was locked before), you will be able to "Claim" it. By claiming, you will get the GFI directly transferred to your wallet.

Understanding the GFI summary

The box on top of the **GFI** tab shows a summary of your GFI distributions.

- **Claimable:** This is the amount of your GFI distributions that you can claim immediately. Claiming transfers this GFI amount to your wallet.

- **Still Locked:** This is the amount of your GFI distributions that is still locked. This amount will unlock according to the unlock schedule. Once the GFI unlocks, it will be claimable.
- **Total balance:** This sum of your claimable and locked GFI.

Understanding a particular GFI distribution

- **Locked GFI:** The amount of GFI that is still locked.
- **Claimable GFI:** The amount of GFI that can be claimed immediately.
- **Transaction details:** A description of how or why you received the distribution.
- **Unlock schedule:** A description of when the GFI will become claimable.
- **Unlock status:** The percentage of the GFI that has already unlocked.
- **Current earn rate:** This only applies to [Liquidity Mining](#) distributions. This is an estimate of how much GFI you will receive per week. This number changes consistently, depending on the size and utilization of the Senior Pool. See the [Liquidity Mining page](#) to learn more.
- **Claim status:** The amount of the GFI you have already claimed.

Accessing Borrower Communication Channels

Step-by-step instructions to access Goldfinch's Borrower communication channels

To learn more about Goldfinch Borrower communication channels, read the [Borrower communication](#) section of the documentation.

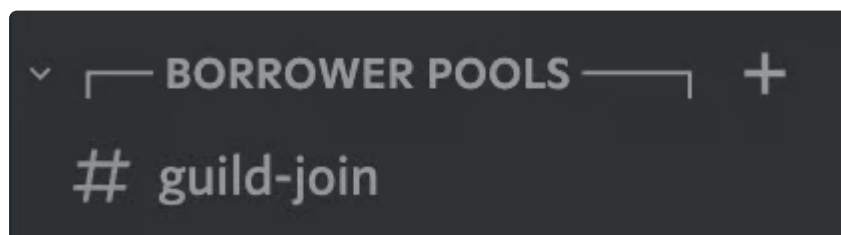
Token-gated Discord channel

As a reminder, these channels are configured to allow only:

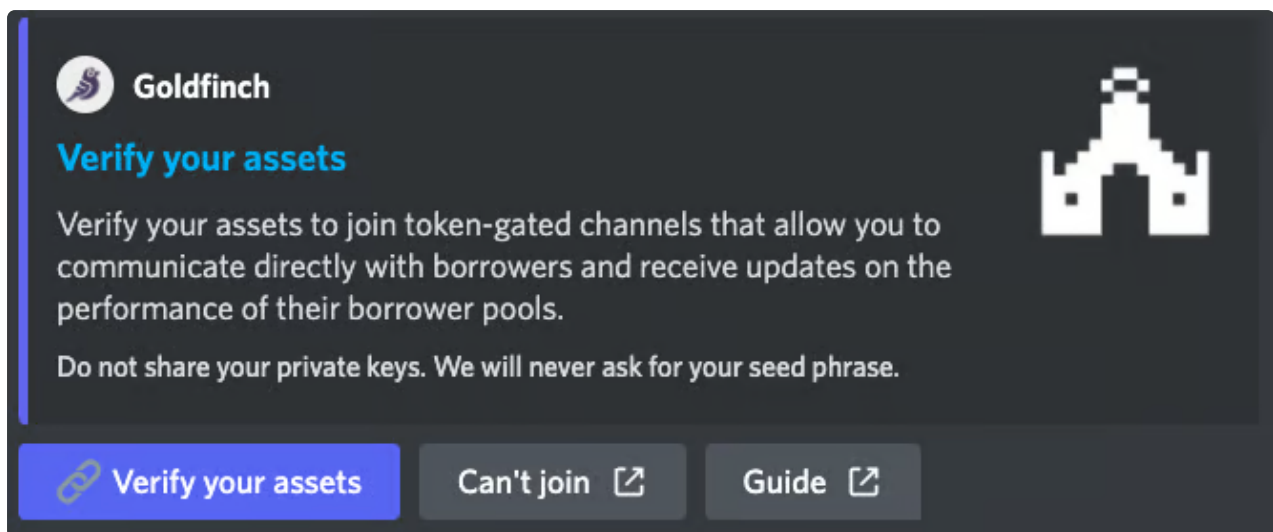
- Backers in the Pool in question, i.e. anyone who has at least one pool token which represents their position as a Backer in the pool, and
- Senior Pool LPs with at least 10,000 FIDU or staked FIDU tokens.
- To protect Borrowers confidential information, Investors will have their access revoked if they don't meet either of the above requirements.

To access the channel:

1. Click this [link ↗](#) to enter the Goldfinch Discord server
2. Once you are in the server, enter the **#guild-join** channel to verify your assets. You will find this channel under the 'Borrower Pools' category.



3. In the channel, you will see the message pictured below. Click '**Verify your assets**' and follow the instructions to verify your assets. Please make sure you are using the wallet that contains your pool tokens to verify your assets.



4. Once your asset verification is complete, you will be able to access the token-gated channels for any pools for which you qualify.

Password-gated Dataroom

In each of these Discord channels, there will be a pinned message containing directions for accessing the password-protected Datarooms where Borrowers will share NDA-protected information. That way, any Backer or LP who accesses the Discord channel using their pool token or FIDU as verification will be able to easily access the Dataroom password as well.

Purchasing coverage with Nexus Mutual

Overview

Buying coverage for your productive assets is the best way to hedge against smart contract risk and protect your funds when you are participating in the Goldfinch Protocol.

Buying coverage for the Goldfinch protocol via industry-leading [Nexus Mutual ↗](#) will provide insurance against your supply of USDC to the protocol's Senior Pool or Borrower Pools in the unlikely event of a smart contract bug resulting in lost funds.

It's worth noting that coverage only covers smart contract risk, and does not apply to the credit risk of Borrowers not paying back their loans.

If you need any support purchasing cover get in touch with the Nexus Mutual team via their [Discord ↗](#) server, on [Twitter ↗](#) or on [Telegram ↗](#).

Factors to Consider when Buying Coverage

Once you know what coverage product you want, you should consider the following factors:

- **How long do you need to be covered?** The minimum duration of a cover policy is 30 days and the maximum duration is 365 days.
- **What payout denomination works best for you?** When you buy your cover policy, you have to select the currency you want to receive a payout in should a loss event occur. You can select ETH or DAI. Most users select ETH as they payout denomination for more volatile assets and DAI as the payout denomination for more stable assets.
- **Which currency do you want to pay for your policy with?** You can pay with ETH, DAI or NXM when you buy your cover policy. Consider which one you want to use for payment.

How to Buy Coverage

Now that you are ready to buy coverage, head to [https://app.nexusmutual.io/home ↗](https://app.nexusmutual.io/home).

On the Home page, you will see the **Welcome** menu below, which displays your current holdings, covers, claims, and membership status. [The view on the Nexus Mutual Home page.](#) ↗

From here, you will select the Buy Cover tab on the main menu. Once you have clicked on that tab, you will see the **Buy Cover** page, which is presented here: [The view on the Buy Cover page of the Nexus Mutual application.](#) ↗

On this page, you can see the different brief explanations of the mutual's cover products. Below these explanations are the various listed protocols, custodians, and yield token cover products.

You can filter by cover product: Yield Tokens, Protocols, and Custodians. You can also enter in the name of the platform or product you want to buy cover for.

In this tutorial, you will follow the process for buying cover by walking through a **Protocol Cover** policy purchase for Goldfinch Finance.

To find the Goldfinch Finance listing, you can enter the name of the protocol in the Search bar. [Click here to see a screenshot.](#) ↗

After searching for "Goldfinch Finance," we can see the Protocol Cover listing for Goldfinch Finance. Because you are protecting a deposit in Goldfinch Finance, you will select **Get Quote** for that listing. [Click here to see a screenshot.](#) ↗

After you click Get Quote, you will see the Quote Details page which looks like the image pictured above

On the Quote Details screen, you will be presented with several fields, which are pictured in the image above and described below:

1. **Amount:** This field is where you select your payout denomination (ETH or DAI) and the amount of capital that you want to cover. While payout denomination is only in ETH or DAI, you can cover any asset that is deposited within a listed protocol, custodian or yield token cover product.

2. **Period:** This field is where you select the duration of your cover policy. The minimum duration is 30 days and the maximum duration is 365 days. Enter the number of days you want your cover policy to protect your assets for.
3. **Terms and Conditions:** This is a brief review of the terms for Protocol Cover, the support chains for the selected protocol, and the conditions for submitting a claim in the event of a loss of funds. Be sure to read this before proceeding to item #4. The link included at the bottom of the Terms and Conditions is the link to the [Protocol Cover wording ↗](#).
4. **"I have read and accept the terms and conditions" field:** Before you can proceed with your cover purchase, you need to select this after you have read the Terms and Conditions.
5. **Grant TokenController Permission:** If you look at the highlighted item under the **Pay In** field above, ****you will see the option to pay in either DAI or NXM. Here, you have selected NXM, so the protocol needs you to grant permission over your NXM to continue with payment. You can approve the NXM necessary for payment OR you can approve an unlimited amount of NXM for this and future payments.
6. **Buy Cover:** After you have submitted the Grant TokenController Permission approval transaction, you will be able to purchase cover. When your approval transaction is complete, you will see this button appear as bright green instead of the dull green pictured above.

Since you have read through the Terms and Conditions, clicked that you have read and agreed to the terms, and have chosen to approve 0.13 NXM to pay for the cover policy, you will be prompted to approve the transaction through your wallet provider. In this case, you are using MetaMask and you see the following: [The approval transaction prompt to grant the TokenController contract permission to spend 0.13 NXM. ↗](#)

To submit the transaction, select **Confirm**.

After you have submitted the approval transaction, you can see the pending transaction in the menu on the top bar of the application: [Set allowance for TokenController transaction is shown as pending in this drop down menu. ↗](#)

Once the approval transaction is complete, you will be able to click **Buy Cover** and move through the prompt for the second transaction, which will execute the purchase of your cover policy. To buy your cover, you must see the following fields pictured here: [Agreed to terms and conditions; granted TokenController permission to spend NXM; ready to buy cover ↗](#).

After clicking **Buy Cover**, you will be asked to approve a transaction. After you submit the transaction, you can see your pending transaction in the drop down menu on the top bar of the application: [Buy Cover transaction is shown as pending in this drop down menu. ↗](#)

Once the transaction is approved, you will have successfully purchased a cover policy from Nexus Mutual! In this case, you purchased a Protocol Cover policy for Goldfinch Finance.

To review your newly purchased cover policy, you can go back to the **Home tab** where you started and look at the **Welcome menu** under **My Covers**. Click here for a [screenshot ↗](#).

You can confirm a cover purchase has been successful by checking the **My Covers** section on this page. Here, you will see a full list of your current cover policies along with the Cover ID, Project Name, Cover Amount, Premium Paid, and the ability to file a Claim.

Now you are ready to buy your own cover policy for the Goldfinch protocol!

Borrower Mechanics

A guide to Goldfinch for Borrowers

 **Goldfinch provides tools to help participants interact with the protocol, but it is your responsibility to stay compliant with all legal, regulatory, and tax laws for your relevant jurisdiction(s). All protocol participants are strongly encouraged to consult with their own legal and tax counsel.**

The Goldfinch protocol helps to simplify debt capital aggregation by operating on the principle of 'trust through consensus. The protocol provides investors around the world the ability to earn yield uncorrelated to crypto markets by helping them invest in:

- Curated real-world debt opportunities from around the world which are assessed by the fellow community members
- On-chain tokenized assets, allowing for secondary markets to flourish
- Democratizing access to yield but allowing anyone to invest with no minimums

You can learn more about Goldfinch and how it works in the [Goldfinch Overview](#) section of the documentation. Then, dive into the [Protocol Mechanics](#) sections of the documentation to learn in more detail about how it functions, along with the [role that Borrowers play in the network](#).

Raising Capital through Goldfinch

Raising capital through the Goldfinch ecosystem can be summarized in four basic steps, which you can learn about in more detail in the [Protocol Mechanics sections](#) of the documentation.

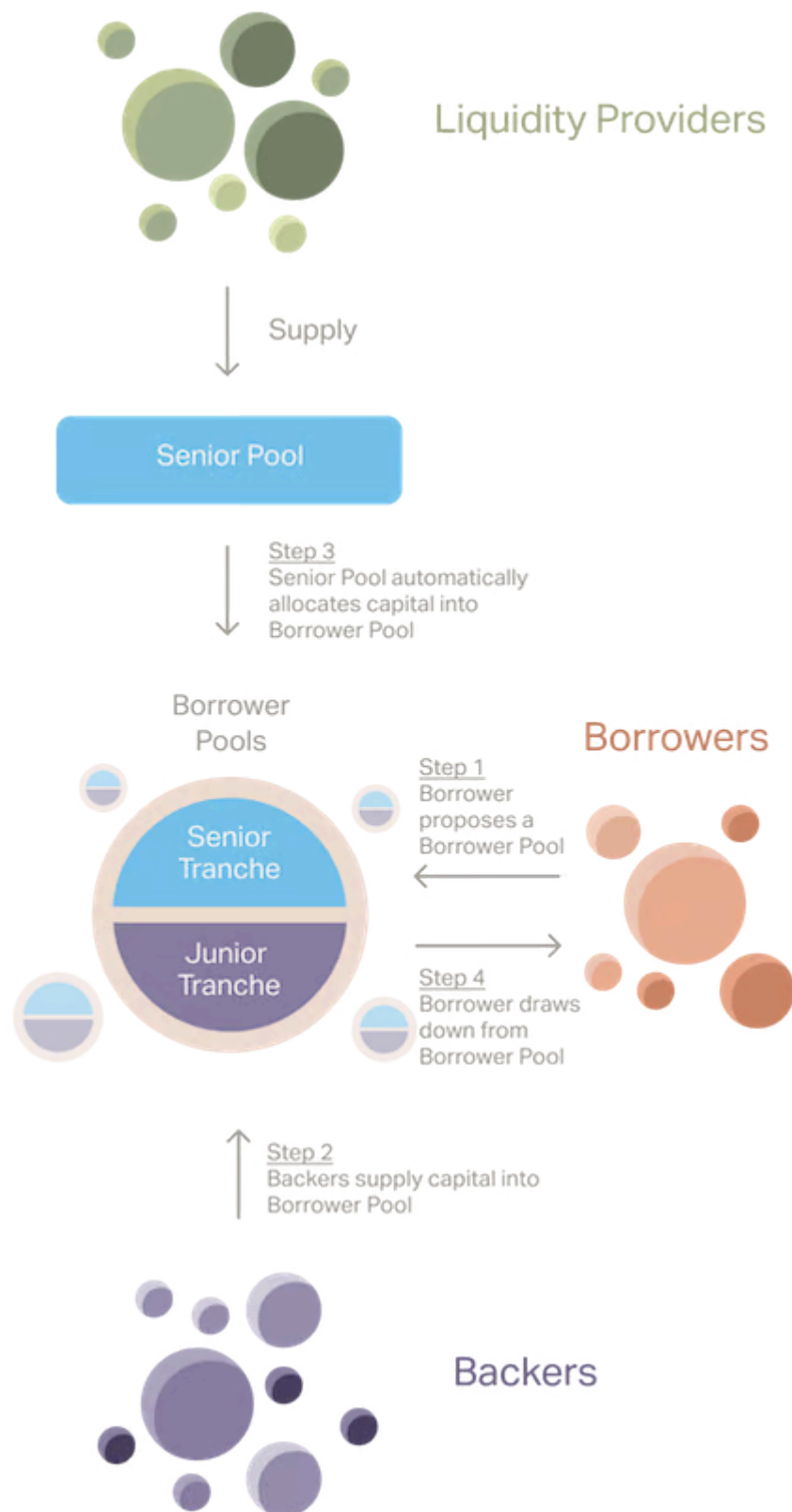
Step 1: A Borrower proposes a [Borrower Pool](#) to the Goldfinch community, sharing deal terms, structure, and due diligence materials with the [Backers](#). Borrower Pools are specific to individual Borrowers, representing the credit lines from which Borrowers draw capital to funds their real-world lending. Each pool is created by the Borrower encoded with a set of deal terms (including pool size, interest rates, repayment schedule etc.) that match the loan that Borrower Pool represents.

Step 2: The Backers then perform due diligence on the Pool opportunity. This diligence is based on the information shared, directly communicating with the Borrower, and negotiating key terms of the transaction. Backers then supply capital once a transaction agreement has been signed by both the Backers and the Borrower.

Step 3: The [Senior Pool](#) automatically allocates its capital to fill deal tickets. Senior Pool capital contribution to the deal based on the [Leverage Ratio calculation](#), which follows the actions of Backers—to date leverage is set at 3x and in future will be determined automatically by the protocol.

Step 4: The borrower draws the USDC raised from the protocol as a unitranche facility.

These four steps typically take a week to complete.



Goldfinch's Capital Providers

There are two types of capital providers on the Goldfinch protocol. They are:

- [Liquidity Providers](#) who supply capital into the Senior Pool, and
- [Backers](#) who supply capital into individual Borrower Pools.

You can learn more about these roles in the [Protocol Mechanics](#) sections of the documentation.

At a high level:

- The Senior Pool is a 'blind pool' of second-loss capital that is diversified across Borrower Pools on the Goldfinch protocol. This means that Liquidity Providers (LPs) who provide capital into the Senior Pool are capital providers in search of diversified exposure across all Borrowers, and do not want to take on individual Borrower risk.
- In contrast to LPs, Backers are active capital providers. They carry out in-depth due diligence on individual Borrowers, and ultimately 'Back' these Borrowers by investing first-loss USDC in the Borrower's Pool. The more Backers who 'Back' the pool, the more trustworthy the protocol believes the Borrower to be.

Sample Backer Economics

Let's take the below deal terms as an example of how back-leverage works:

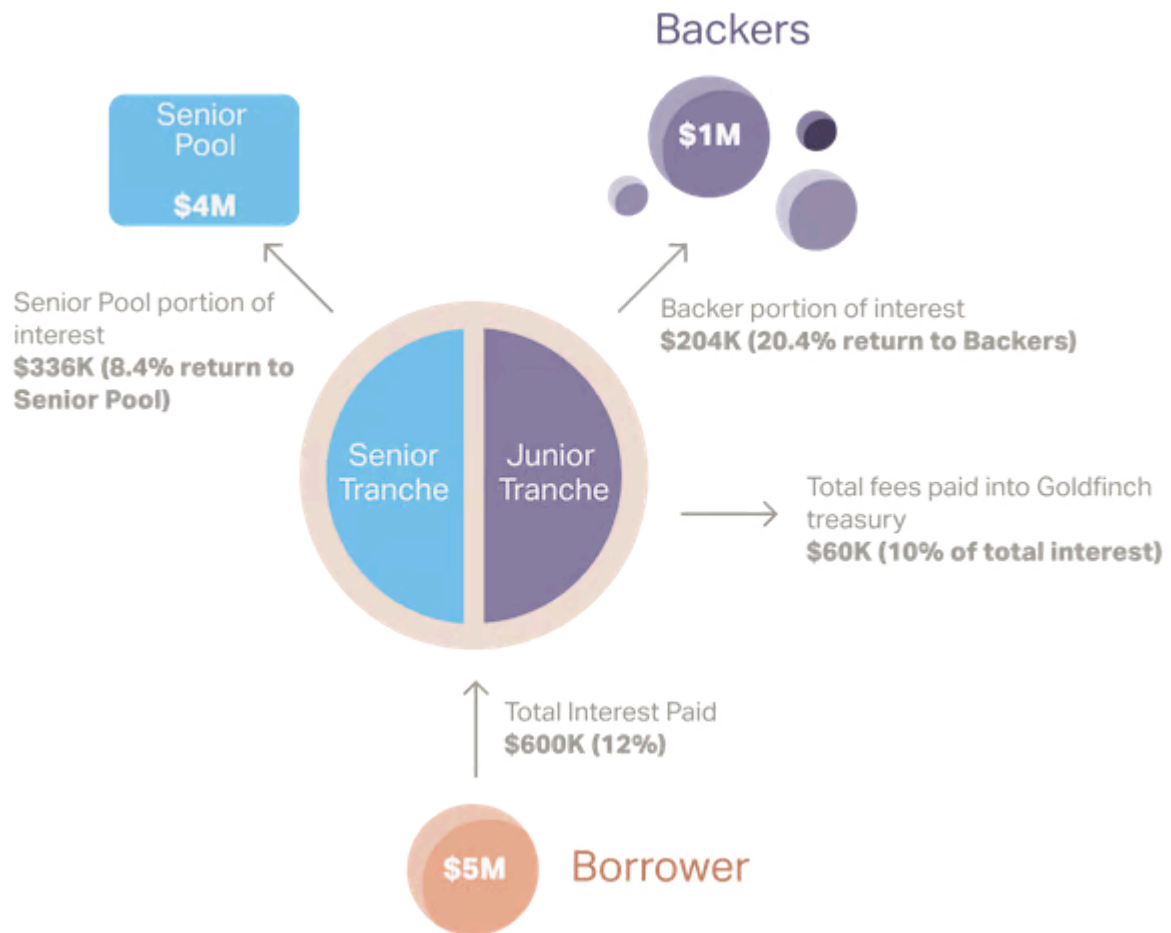
Hypothetical Transaction:

- Facility Size - \$5,000,000
- Borrower Coupon - 12%
 - Senior Net Coupon = % Gross Coupon - 20% Junior Relocation - 10% Pool Reserve
 - Junior Net Coupon = % Gross Coupon + 20% Junior Relocation - 10% Pool Reserve
- Leverage Ratio - 4x
- Backer Investment: \$1,000,000
- Senior Pool Investment: \$4,000,000

Senior Pool Economics: 8.4%

Backer Economics: 20.4%

The table below shows the economics for both the Backers and the Senior Pool for a Borrower Pool with details above. You can download this sheet [here ↗](#).



Goldfinch's Borrowers

You can learn more about Borrowers and the role they play in the protocol in the [Borrowers](#) section of the Protocol Mechanics documentation.

- Borrowers are able to raise unitranche loan structure allowing them to pay a single interest rate with all interest and principal payments made into a single portal
- Borrowers have the ability to propose any investments based on any terms, and transaction structures to the Goldfinch community
- The Goldfinch ecosystem was built to be generalized allowing for many Borrower archetypes that could raise capital through it. Some examples Borrower types include, but are not limited to:
 - Credit Funds
 - Fintechs

Below are some articles highlighting past borrowers on the protocol:

- [Goldfinch Provides Cauris with \\$5.1M as Part of a \\$100M DeFi Fintech Debt Fund ↗](#)
- [Goldfinch Borrower Spotlight: Greenway \(via Almavest\) ↗](#)
- [Goldfinch provides Tugende with a \\$5M debt facility to scale its operations in Kenya ↗](#)

Template Borrower Deal Structures

Here are a few simple archetypes of structures, Borrowers such as Credit Funds and Fintechs could design and propose to the Goldfinch community.

Key Universal Structural Considerations

While there are many use cases for the Goldfinch ecosystem (see below), there are two universal questions which should be carefully considered prior to approaching the Goldfinch community.

1. "How will the transaction security which typically can be assigned to a single party in a transaction be perfected, and accessible to investors?"
2. "Who at the outset of pool close would be the primary delegate acting on the best interest of the other Backers when it comes to managing communications"

Credit Funds

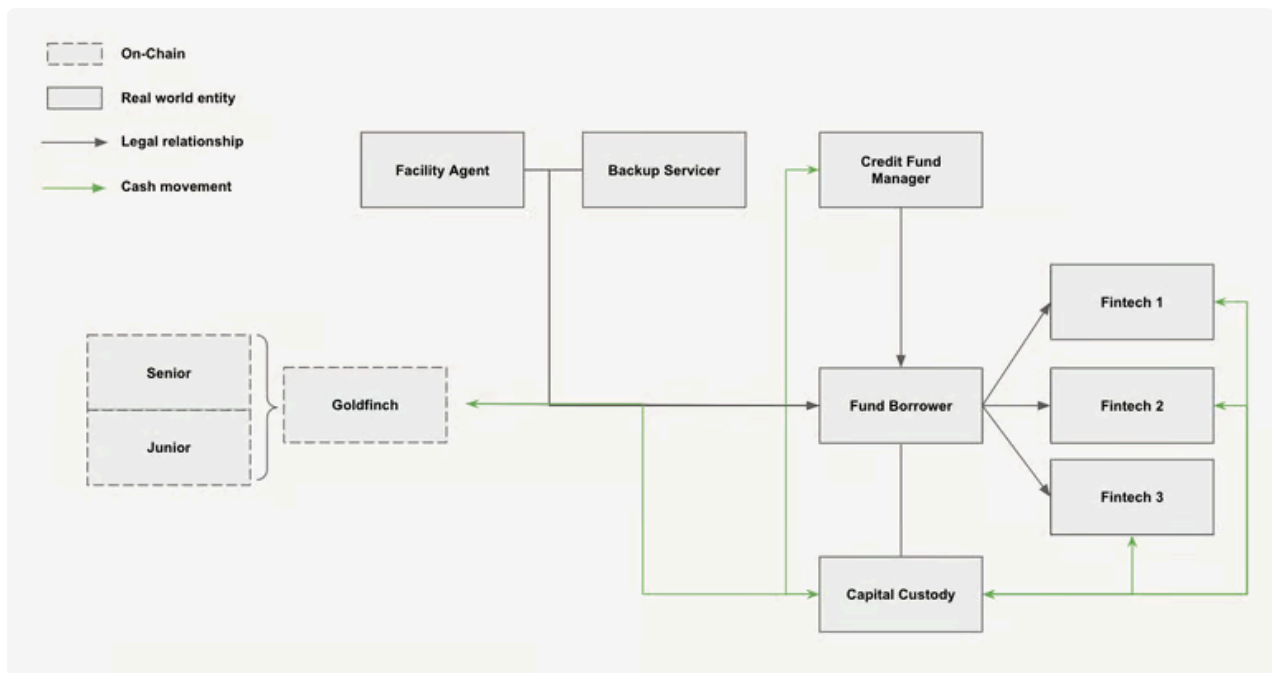
There are a multitude of use cases Credit Funds could deploy capital raised through the Goldfinch ecosystem into. The following illustrates just a few:

- Raise parri-passu or senior capital into your fund (or fund like) structure to then on-lend
- Co-invest into a new prospective borrower via raising new money capital
- Co-invest into an exiting borrower via raising new money capital (i.e. fill uncommitted allocation)
- Refinance existing investments via sub participate exiting investments at a different cost of capital

Checklist of Features Credit Funds Could Incorporate

- ☐ **Facility and Security Agent** - A single party appointed by the other Backers to manage the perfected security, and core communication between the Borrower and investor group

- ☐ **Backup Servicer** - A 3rd party with the ability to step in as the fund administrator in the instance the Credit Fund were to default
- ☐ **Capital Custody** - A capital custody platform(s) to manage Crypto or fiat capital

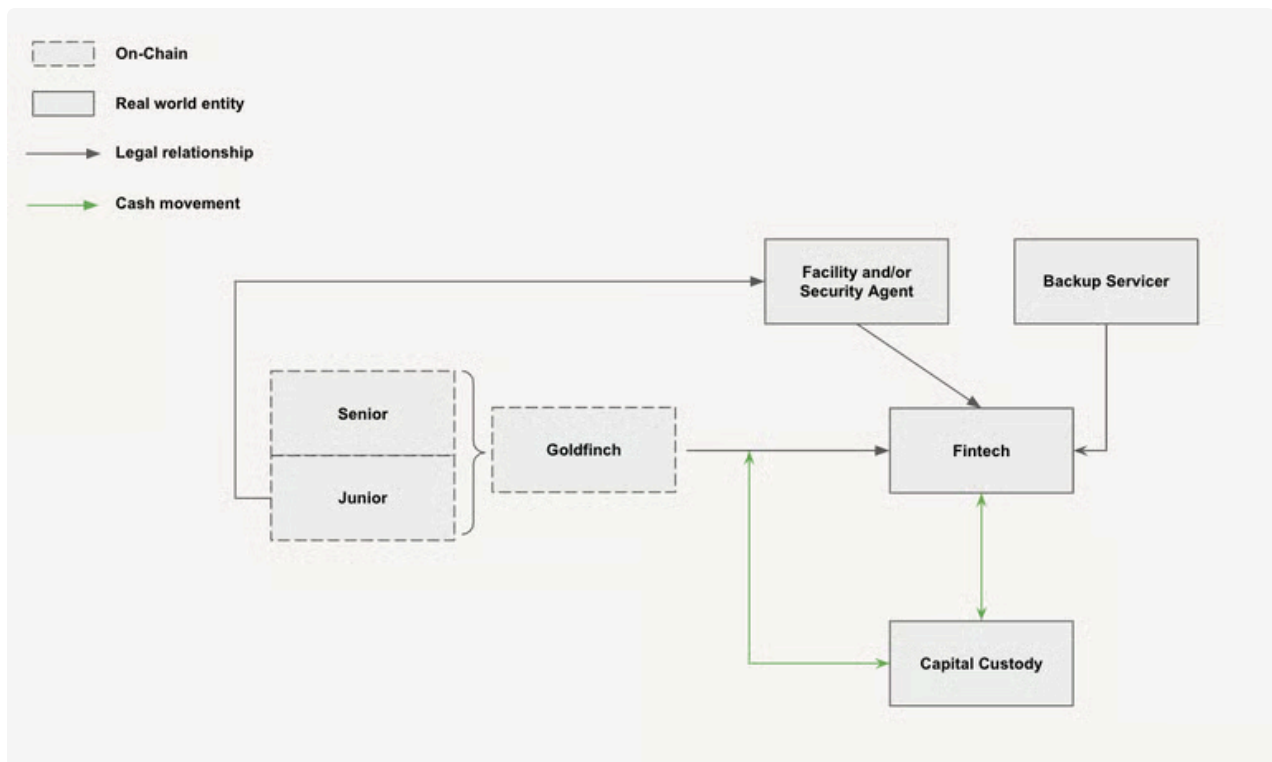


Fintech Structural Diagram

The following is a potential "plug and play" structure speaking to the key universal structural considerations.

Checklist of Features Fintechs Could Incorporate

- ☐ **Facility and Security Agent** - A single party appointed by the other Backers to manage the perfected security, and core communication between the Borrower and investor group
- ☐ **Backup Servicer** - A 3rd party with the ability to step in to collect funds from the Fintechs end borrowers
- ☐ **Capital Custody** - A capital custody platform(s) to manage Crypto or fiat capital



Borrower Case Study

Almavest - August 2021

Fund: [Almavest](#) ↗

Background: Almavest is a credit fund that provides debt capital to fintech lenders, and carbon reduction project developers globally. In August 2021, Almavest launched a Borrower Pool on the protocol to raise ~\$2M. The purpose of this Borrower Pool was to fund a basket of fintech lenders in the Global South.

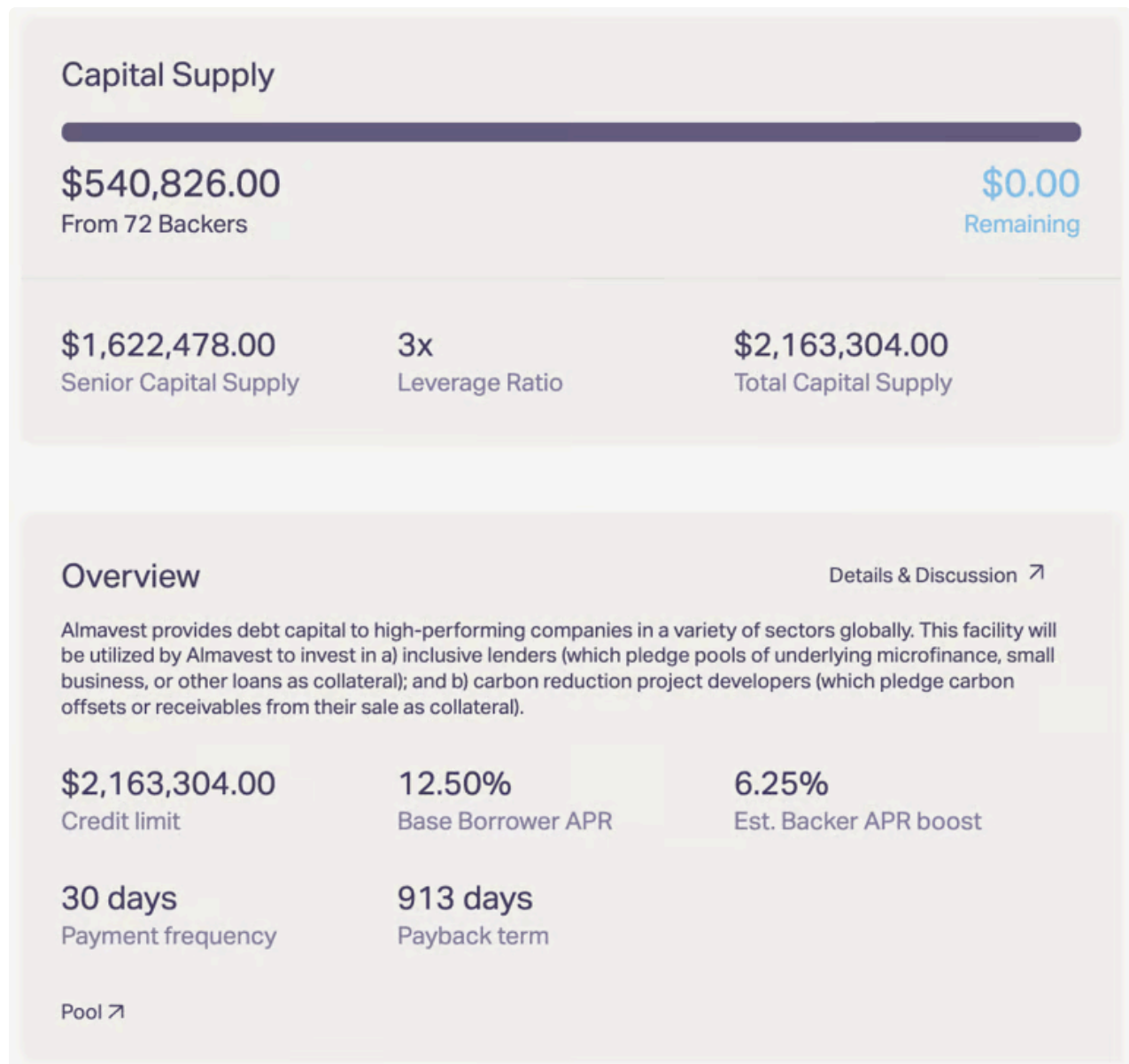
Process: Over one week, the Almavest team underwent due diligence by the Backers, presenting them with a dataroom, and being on hand to answer questions posed by the Backers. This included a live video Q&A session you can find [here](#) ↗.

Once the due diligence period was done, the pool opened on August 31st for funding.

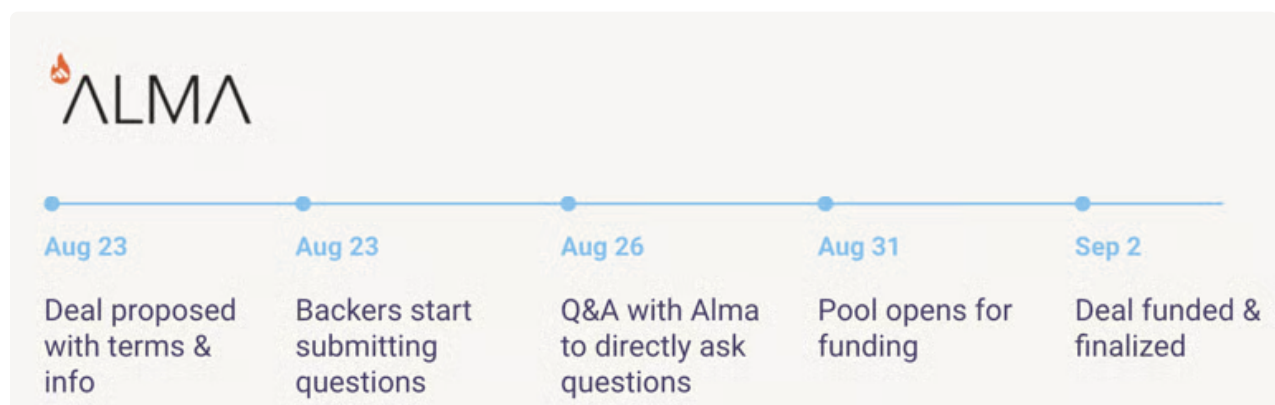
Outcome: The junior portion of the Borrower Pool was filled by 72 Backers, who provided \$540,826. The Senior Pool provided an additional \$1,622,478 (3x the Backer contribution) to complete the \$2,163,304 raise. This entire process was completed in 30 minutes. Below is a screenshot from the Almavest Borrower Pool. You can find the live page [here](#) ↗.

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Almavest Pool Snapshot



Almavest Timeline and Key Dates



- **Announce Investment to Community**
 1. Announce deal to backers with key timelines
- **Deal Proposed With Terms**
 1. Surface NDA, Dataroom, and Summary of Terms Backers are expected to review
- **Backers start submitting questions**
 1. Create a dedicated channel to accept and respond to questions through
- **Q&A with Backers**
 1. Create a curated AMA session
- **Pool opens for funding**
 1. Confirm date you wish to open the pool for Backer funding
- **Deal funded & finalized**
 1. Drawdown capital into wallet, and wire to crypto business account to be converted to fiat ccy
 2. Post final docs up in dataroom for final selected investors

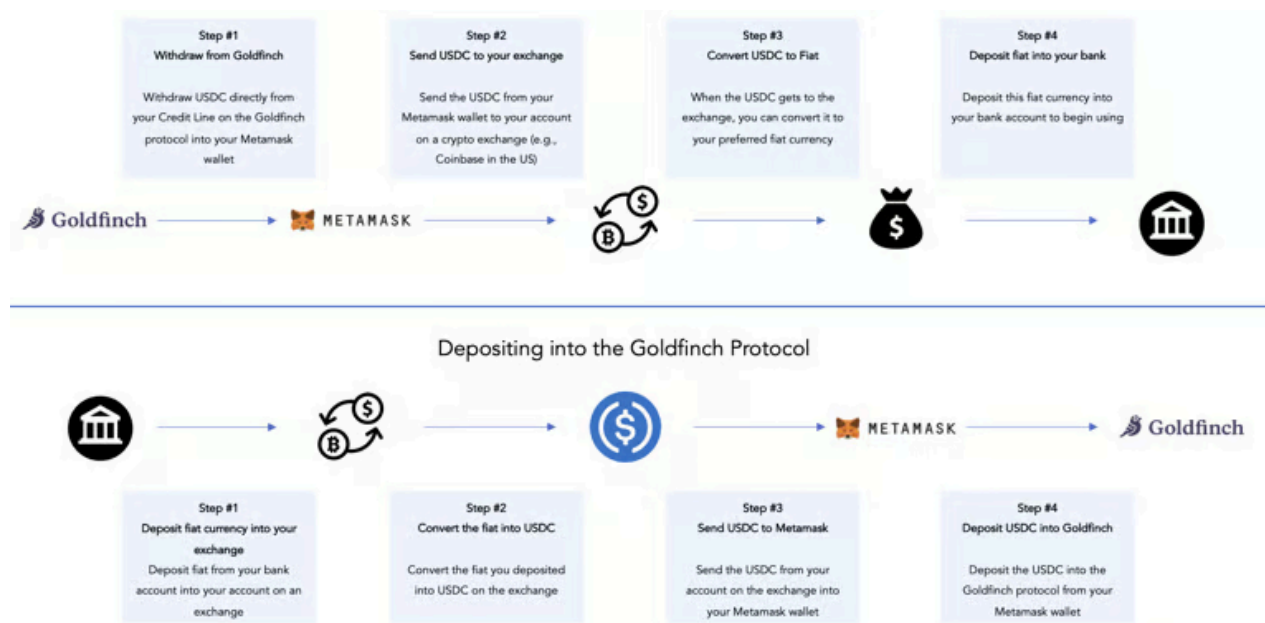
Structure and Legal Considerations

 **Goldfinch provides tools to help participants interact with the protocol, but it is your responsibility to stay compliant with all legal, regulatory, and tax laws for your relevant jurisdiction(s). All protocol participants are strongly encouraged to consult with their own legal and tax counsel.**

Below are some important structure and legal considerations to take into account as you look to get started on the Goldfinch Protocol, including cashing in and out of the ecosystem, Investor KYC/AML requirements, key mechanics to consider including Debt Facility Mechanics and Transaction Documentation, Senior Pool participation for Borrowers, and Backer transferability of debt positions.

Cash in and out of Goldfinch

As a Borrower (or Lead Backer) you will directly interact with the protocol by withdrawing and depositing USDC (a US dollar-backed stablecoin) into the protocol using your [Metamask](#) wallet. Below is a diagram that explains your interaction with the protocol:



Read on for a list of third-party service providers to help facilitate your movement from crypto to fiat and vice versa.

Recommended Service Providers

Recommended Crypto Exchanges

Exchange	Coverage / Resources
Coinbase ↗	https://www.coinbase.com/places ↗
Kraken ↗	https://support.kraken.com/hc/en-us/articles/360001368823-Where-can-I-use-Kraken- ↗
Circle ↗	https://www.circle.com/en/circle-account ↗
Wyre ↗	https://support.sendwyre.com/hc/en-us/articles/360055233754-Geographic-Restrictions- ↗
Independent Reserve ↗	https://www.independentreserve.com/about ↗
Sygnum ↗	https://www.sygnum.com/solutions/ ↗
StraitsX ↗	https://www.straitsx.com/sg/about ↗
Hextrust ↗	https://hextrust.com/company/about-us ↗

Recommended Crypto-Friendly Banks

Bank	Country Coverage
Mercury Bank ↗	USA
Silvergate Bank ↗	USA

Investor KYC / AML Requirements

Investors in the Goldfinch ecosystem are required to mint a [Unique Identity NFT \(UID\)](#) [↗] in order to gain access to Goldfinch pools. The UID represents that an investor has passed through various OFAC, AML, Sanctions checks which are conducted by [Persona](#) [↗]. You can learn more about the need for a [Unique Entity Check on Goldfinch here](#), and about the [Unique Identity NFT process here](#).

Key Mechanics to Consider

The Goldfinch protocol introduces key non-traditional features that Borrowers should be aware of. Below are a few of these features. They are detailed in the following two pages, grouped as follows:

1. Debt Facility Mechanics
2. Transaction Documentation

Debt Facility Mechanics

Below are some key mechanics of a debt facility on the Goldfinch Protocol

Revolving Credit Facility

Borrower Pools on the Goldfinch protocol work like a revolving credit facility. Borrowers have the ability to withdraw and repay (as many times as they like) any amount provided the total outstanding amount at any time is less than the Borrower Pool's credit limit. There are no prepayment penalties.

However, it is worth noting that any amount of capital left unutilized in a Borrower Pool can be withdrawn by Backers at any time. This means that although the protocol offers a revolving credit facility, it is not a committed facility. It is designed this way to give Backers liquidity, and to ensure that borrowers do not pay fees on unused capital)

Facility Tranches

Each borrower pool has the ability to accommodate multiple tranches of the same facility. This allows borrowers to grow the size of capital they draw from the protocol without having to launch new Borrower pools.

Interest Payments

The Goldfinch protocol is built to expect interest payments to be made every thirty days, starting from the day a Borrower draws down funds into their wallet. Interest is calculated on an Actual / 365 basis

Facility and Security Agents

As no centralized party provides any administrative roles for transactions on the Goldfinch protocol, Borrowers need to externally source and bear the cost of any facility/admin/security agents required in agreements.

Transaction Documentation

Below are some key things for Borrowers to note when preparing transaction documentation for a Borrower Pool on the Goldfinch Protocol

Currency Denomination

The Loan Documents should be denominated in USDC (a USD backed stablecoin)

Loan Instrument Documentation

Borrowers should use traditional debt agreements as the starting point for their loan documents. The on-chain / tokenization elements of the transaction happen via the Goldfinch protocol and do not need to form the basis of your loan documentation

Borrower Repayment Clause

The loan documents should dictate that all Borrower payments be made only through the Goldfinch protocol. This is to ensure accurate waterfall calculations between the Backers and the Senior Pool. This waterfall is calculated by the Protocol.

Note that these repayments mechanics are binding, and may not be changed by any party (neither the Borrower nor Backers) in a Borrower Pool. Below is a sample language that may be used in your loan documentation to reflect the above:

Payments. All payments due shall be paid by the Borrower on the Goldfinch Protocol which will in turn distribute the Senior Participant Payment and Lenders payments. Payments should be made by the same wallet that received the funds and be paid to the same wallet that disbursed the funds.

Lender of Record

It is important to note that the sole parties to the final transaction are the Borrower and Backers.

Goldfinch is not a party to any transaction agreement. Backers individually sign and collectively act as the counterparties to the relevant transaction agreements. After the close of each pool, the list of participating Backers shall be provided to the Borrower by the community. This list constitutes digital signatures by each Backer. Also note that if the Borrower or a related entity to the Borrower contributes to the Borrower Pool, they will have to recuse themselves as a lender.

Lender Signatures: In the process of funding a Borrower pool, each Backer electronically signs the loan documents for that pool. Once the pool is fully funded, the community will provide the Borrower with a complete list of Backers along with the time stamp on the Backer's signing. This list is to be appended as an Exhibit to the final loan documents

Lender Decisions: Any and all lender decisions may be communicated via the electronic data room. Along with each communication, evidence must be provided of on-chain agreement amongst lenders (for example, in the form of a governance vote) has been made to serve the notice

Borrower Responsibility: As a Borrower, you are responsible for including all relevant disclosures, representations, and warranties related to your particular jurisdiction, facts, and circumstances in the Loan Documentation. Please also note that Borrowers cannot act as Facility and Security Agents. These will have to be sourced externally

Single Certificates: To the extent that any certificates or other documents have to be created as part of the facility, they should be pledged to the Lenders collectively (i.e. a single document). The Goldfinch protocol will allocate the rights automatically to each Lender on a pro-rata basis.

Pro-rata rights to Lenders: Please note that if for any reason the debenture amount does not equal the facility amount, rights are allocated on a pro-rata basis to the Lenders

Backer Transferability

Restrictions on a lender's ability to transfer their positions must be removed from all loan documentation. The Protocol will update borrowers as to changes in their lenders. Please note that any redemption interest of principal repayment of each loan can only be made by a lender who is KYC'd

Repayment Accounting and Documentation

Interest on the Goldfinch protocol is calculated every thirty days. Traditional clauses that otherwise define the repayment schedule, such as adjustment for business days, must be changed to reflect the protocol's payment mechanism. Below is a sample catch-all phrase that Borrowers can use, for an abundance of clarity:

Notwithstanding any other provision in this Agreement, the Parties acknowledge and agree that the amount of interest payable by the Issuer to the Lenders and the date on which such payment is due shall be subject to the schedule set out in the Goldfinch Protocol, which absent manifest error, shall supersede in the case of any inconsistency with Exhibit "B".

Repayment Accounting and Documentation

Borrowers are required to provide covenant compliance certificates to the Lenders at a predetermined frequency. This requirement is to be included in the loan documentation.

Tax gross-up

Borrowers must gross-up taxes, including withholding tax payments. The Protocol must receive a fixed amount of interest specified for the loan. This is to be reflected in the Loan Documentation

Electronic Delivery of Documents

All documents and notices must be delivered in digital form and must not require wet-ink signatures or physical receipt. This applies to documents from both Borrower and Lenders and includes, amongst others,;

- Executed documents related to the loan
- Any notices to enforce rights
- Any reports required per document reporting requirements

Other Important Definitions

See below other definitions to be included in the transaction documentation:

- "Goldfinch Protocol", means the Goldfinch protocol, a decentralized crypto borrowing platform found at the following website <https://app.goldfinch.finance/earn>.
- "USDC" means is a tokenized U.S. dollar, for which the value of one USDC coin pegged 1:1 to the value of one U.S. dollar.
- "Unique Identifier" or "UID" means an NFT that represents an entities KYC status that enables them to participate on the Goldfinch protocol

Borrower Senior Pool Participation

It is important to note that neither the Senior Pool nor the investors in the Senior Pool are lenders of record. The Senior Pool participates in the economics of the transaction by virtue of the protocol design (i.e. the smart contract).

All Senior Pool investors sign up to the [following agreement ↗](#), covering Reg D and non-US requirements, ahead of being able to access the Goldfinch protocol.

 Borrowers should consult with their own legal, tax, and regulatory counsel to review the above agreement in relation to other documents executed by Backers in their raise

Backer Transferability

NFTs as a representation of Backer Participation

Once a Backer invests into a pool on the Goldfinch protocol, they receive a non-fungible token (NFT) that represents their investment into the pool. You can learn more about the function of that NFT in the [Backers section](#) of the documentation. It is important to note that this NFT could be freely transferable to any other Backer, such as by the implementation of backer secondary markets as outlined in the [protocol's community-driven roadmap](#) ↗.

This could mean the Backers in your Borrower Pool could change during the life of the loan (e.g. backers trade their loans), although interacting with the Borrower Pool smart contracts will still have the same [UID requirements](#).

Borrower How-To

Educational guides on how to use the protocol, for Goldfinch Borrowers

Launching a Borrower Pool on Goldfinch

As you look to launch a borrower pool on the Goldfinch protocol, here are the operational considerations you should take into account. For simplicity, we have broken this down into stages. We recommend budgeting a minimum two months of operational prep ahead of your initial launch on the platform.

Currently, as a first step anyone who is interested in Borrowing on Goldfinch should fill out the [Borrower Inquiry form ↗](#).

 **Goldfinch provides tools to help participants interact with the protocol, but it is your responsibility to stay compliant with all legal, regulatory, and tax laws for your relevant jurisdiction(s). All protocol participants are strongly encouraged to consult with their own legal and tax counsel.**

Stage 0: Preliminary Actions and Decisions

Below are some of the early actions you will need to take when exploring a deal on the Goldfinch protocol:

- **Create your Metamask Wallet:** As described above, you will need a Metamask wallet to interact with the Goldfinch Protocol. You should set this up as soon as possible. You can begin the process of setting up your wallet [here](#) ↗
- **Open your Crypto Exchange Account:** You will also need to set up your business account at a crypto exchange to help you convert from fiat to crypto and vice versa. This can be a time-consuming process, so we advise you to get started on this as soon as possible. [Here](#) are some crypto exchanges we would recommend. Please let us know if you would like introductions to any of the below
- **Open a Bank Account with a Crypto-Friendly Bank:** Please begin the process of opening a bank account with a crypto-friendly bank. As you will be receiving large sums of money from your crypto exchange account, it is important to work with a bank that is amenable to crypto transactions. Please see below a list of banks that we would recommend
- **Familiarize yourself with communications platforms** like Discord, Telegram, Signal, etc. which you need to communicate with the Goldfinch community during your raise process. You can learn more about these [here](#).
- **Counsel:** Select counsel which is familiar with securities law in the jurisdiction into which you are raising. Your counsel would be responsible for all securities and disclosure advice. Please refer to the KYC Requirements in the Legal Considerations mentioned above for context.

Stage 1: Deal Structure and Timelines

Here, you will need to decide on the structure and terms of the deal in question, as well as the timeline for bringing the borrower pool to the community and ultimately getting it funded

Here are some recommended sample deal structures

Stage 2: Dataroom Preparation

In preparation for Backer due diligence, you will need to stand up a virtual dataroom. For this, we recommend you use a virtual dataroom service that allows you to track investor engagement (e.g. [Docsend](#) ↗).

At the minimum, your dataroom should include:

- An overview speaking to your historical performance
- Material speaking to the performance of your borrowers
- Transaction structure and security overview document explaining how security works in the instance of a default
- Agreements you have in place with your borrowers
- Deal Specific Documentation such as:
 - Term sheets
 - Loan Agreements
 - Security Agreements
 - Any other transaction documents you think necessary

Stage 3: Announcements and Tooling Set Up

As we get closer to going live, you will need to decide on what tools to use to run the funding process. These include tools for:

- Sharing and signing of a **Non-Disclosure Agreement** between your fund and the Backers who will be performing due diligence. We recommend that this be a one-way NDA
- A **Dataroom** to share diligence information between you and the Backer community. This can be hosted on tools similar to [Docsend ↗](#) or [Box ↗](#) allowing you to provide tracked, tiered access to your data
- **Two-way messaging** between you and the Backer community. This can be hosted on a bilateral communication tool allowing you to respond to communicate with Backers. Examples of such tools are [Discord ↗](#) or [Telegram ↗](#). Here are descriptions of how to set up a [Discord server ↗](#), and a [Telegram group ↗](#) respectively.
- An optional **video AMA session** where the Backers get to learn about your opportunity and ask you questions. You can host this on platforms such as [Youtube ↗](#) or [Crowdcast ↗](#) to help add a human touch to your raise
- Optional **co-marketing** on the launch of your pool with the Warbler Labs team

You will also have to decide on the content and timing of announcements that go out to the Goldfinch community.

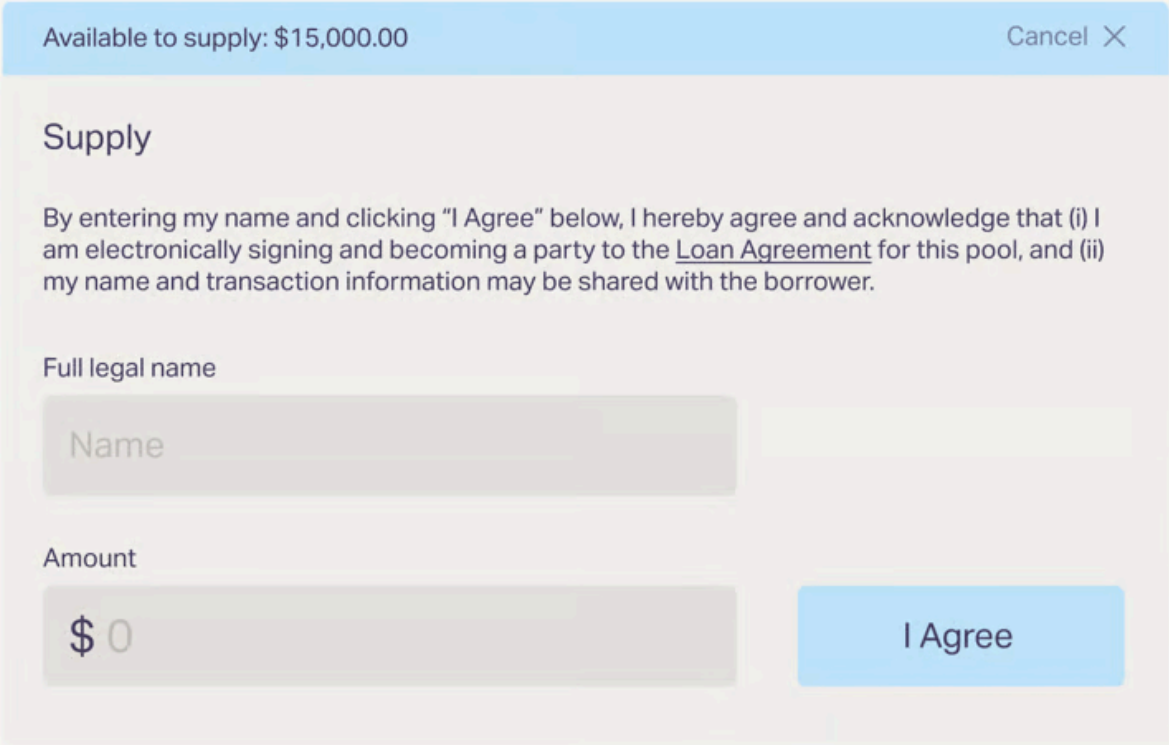
Below is a template document to help you track of all things tooling and announcements

[Template Announcements + Tooling ↗](#)

Stage 4: Borrower Closing

Once all the setup for the Borrower Pool is done, it's time to open the borrower pool and close the fundraiser. This happens in the below steps:

- **Open the Borrower Pool:** On the pre-agreed day, the Borrower pool will open for funding
- **Backer Funding:** Once the Borrower pool opens, Backers will be able to fund your Borrower Pool. They do so by:
- **Signing the Agreement + Funding** - Below is an image of the Agreement Signing and Funding interaction



The screenshot shows a web interface for a Borrower Pool. At the top, a blue header bar displays "Available to supply: \$15,000.00" on the left and a "Cancel X" button on the right. Below this, the section is titled "Supply". A paragraph of text states: "By entering my name and clicking 'I Agree' below, I hereby agree and acknowledge that (i) I am electronically signing and becoming a party to the [Loan Agreement](#) for this pool, and (ii) my name and transaction information may be shared with the borrower." Below the text, there are two input fields: "Full legal name" with a placeholder "Name" and "Amount" with a placeholder "\$ 0". To the right of the "Amount" field is a blue button labeled "I Agree".

Please note that the 'Loan Agreement' hyperlink above takes the Backer to a copy of the Loan Agreement that the Borrower provides.

- **Funding Complete:** Once the pool has been fully funded, the Borrower Pool will be locked

- **Borrower Executes the Agreement:** You will receive the details of the Backers who participated in your Borrower Pool. You will be responsible for appending their details to the loan agreement and executing the agreement.
- **Creating a live dataroom:** You will be responsible for creating a live dataroom, and granting access to all Backers who participated in your Borrower Pool. In this dataroom you will upload:
 - The executed transaction agreements
 - Any relevant updates for the Backer community from time to time.

General FAQ

How do I get involved in the Goldfinch community?

There are several ways to get involved in the Goldfinch protocol. The best ways to get involved in the protocol are:

- Participating as an active community member by joining the [Discord](#). ↗
- Being a Liquidity Provider in the Senior Pool (see [Liquidity Providers](#) to learn more).
- Being a Backer to Borrower Pools (see [Borrower Pools](#) to learn more).

Do I need to be an Accredited Investor to participate in the Senior Pool?

US residents must be Accredited Investors to supply capital to the Senior Pool. Anyone outside of the US within regions supported by UID can participate in the [Senior Pool](#). To learn how to participate, click [here](#).

Where can I find the updates of the Goldfinch Protocol?

You can sign up for weekly Goldfinch Updates on the [Goldfinch Substack](#) ↗, or follow along live on [Twitter](#) ↗ and [Medium](#) ↗. You can connect with the community to stay connected with what's going on in the [Goldfinch Discord](#) ↗ server and [Telegram](#) ↗. To receive live updates when new Borrower Pools are launching, sign up for [Backer Updates](#) ↗.

I don't speak English. Will Goldfinch dapp be available in my language?

The app is still in English. However, there are different language channels set up in Discord - Russian, Ukrainian, Korean, Turkish, Indonesian, Vietnamese, Spanish, among others. These channels are great for asking any questions, and whenever an announcement goes out in English, these channels share a translated version. Community members have volunteered to translate the content into different languages, and any member should feel empowered to [request](#) ↗ to have one, and the community moderators will set it up if there is a high need.

Is Goldfinch a blockchain?

No, it is not. Goldfinch is built on top of the Ethereum blockchain.

Is Goldfinch open source?

Yes. All code, including the front-ends can be found at <https://github.com/goldfinch-eng/mono> ↗. Just the smart contracts and associated audits can be found here: <https://github.com/goldfinch-eng/goldfinch-contracts> ↗.

Does Goldfinch have a Bug Bounty program?

Yes, Goldfinch has a [Bug Bounty program](#) ↗ in partnership with Immunefi.

Which wallets does the dapp support?

The dapp supports [WalletConnect](#) ↗, [Metamask](#) ↗, and [Ledger](#) ↗. Ledger performs optimally through [Firefox](#) ↗. [Trezor](#) ↗ is currently not supported.

What is the currency of the loans?

All loans are in USDC. You can learn more about USDC [here](#) ↗.

Does Goldfinch really take no collateral in its deals?

Goldfinch does not ask Borrowers to deposit any on-chain collateral to take out a loan. Instead, when structuring a loan, Borrowers pledge real-world assets as collateral.

Has the Goldfinch protocol been audited?

Yes, by [Certik](#) ↗ and [Trail of Bits](#) ↗. The initial V2 of the protocol (the design outlined in the white paper) has been audited by Certik (here is the [Github repo](#) ↗). Another audit was done by Trail of Bits (<https://www.trailofbits.com/>) in November 2021 (here is the [Github repo](#) ↗).

Who are the current Borrowers using the protocol?

The list of past borrowers is available publicly [here ↗](#) under `Borrower Pools`. The focus right now is on credit funds and later stage borrowers who are series B and beyond.

I heard about "Flight Academy," what was that?

Flight Academy was a program that empowered the Backer community with tools and basic vocabulary needed to evaluate Debt Capital Markets deals. The program aimed to establish community values and set community norms. The academy ran from September 20, 2021, to October 25, 2021. The academy tutorials are [available ↗](#) for free. Participants were rewarded with GFI (Goldfinch's token) based on a specific criteria, as explained in the final announcements [here ↗](#), and [here ↗](#).

Currently, the community has not proposed hosting another Flight Academy. However, community members should feel empowered to propose one through governance.

Backer Incentives vs. LP Incentives

There have been a lot of questions about the fact that retroactive distributions happened on launch day (Jan 11th) for LPs and not for Backers. Per the white paper, and because of this community-approved [governance proposal ↗](#), Backers will now receive distributions for participating in Borrower Pools. The smart contracts for Backer distributions have already been audited and deployed, and are already on the community Github repo [here ↗](#).

Read more about [Backer Incentives here](#).

If a Borrower defaults, is the money paid back on-chain or off-chain?

In practice, there are security agents with each Borrower that are instructed to liquidate the pledged collateral, convert into USDC, and make payments on chain into the relevant Borrower pool. The Senior Pool and Backers are then able to claim their portion of the funds from the Borrower pool.

What sort of deals can the Goldfinch protocol be used for?

Almost any kind of credit-related investment, into a real-world venture providing some sort of productive service. The Goldfinch protocol is a marketplace, where capital is aggregated for various use cases as it relates to lending. The transaction sponsor (e.g. borrower, or lead backer) owns presenting an opportunity to the community (e.g. principal investment into a corporate, credit fund co-investment, credit fund sub-participation etc...), with the community, ultimately deciding whether or not to fund the opportunity.

What type of structured facility can the Goldfinch protocol provide Borrowers?

The protocol provides a uni-tranche facility, whereby a single credit line is made available, at a single coupon. There is no ability to only raise junior, or senior capital through the Goldfinch protocol.

How can I assess whether the Smart Contract which governs how capital moves works?

The smart contract code is freely accessible and can be found in Github. The Goldfinch smart contract has additionally been audited by 3rd party providers for security flaws by teams like Trail of Bits. You can learn more about the smart contract and Goldfinch's code in the [Developer Documentation](#).

Bots are withdrawing USDC from the Senior Pool as new money comes in—can you stop this? What's going on?

In times of very low excess liquidity in the Senior Pool, LP holders are unable to withdraw USDC from the Pool directly until new repayments are made. Instead, they can turn to secondary markets (like [Curve](#)) to sell their position. This can mean FIDU sells at a discount on those markets until more liquidity comes into the pool.

This discount creates an opportunity for arbitrage bots to pull out the new deposits and repayments as soon as they come in (by buying discounted FIDU on the secondary market, and selling it to the Pool at the higher, native price). If you're an LP who isn't running a bot, and you want to exit, this can be frustrating, and so some people have asked that the community to stop the bots in some way, thinking that this may stop the issue of low liquidity.

While developing certain new mechanics could help, it's important to remember 1) the bots are **an effect, not a cause, of low liquidity**, as low liquidity drives the discount, and 2) no one can really stop someone from running a bot. So, stopping them wouldn't really change the situation.

The secondary markets give LPs with the most need for liquidity the ability to exit now (at the expense of slippage), even when "native" liquidity isn't available in the Senior Pool.

The secondary markets do not fundamentally change the withdraw pressure. They just create a market pricing mechanism for whoever is willing to "pay the most" for the ability to exit now.

While eventually Borrower repayments will cover everyone in the Senior Pool, the best overall solution is to continue to drive new liquidity to the Pool. Once the arb opportunity closes, then any new liquidity will stay there and arb bots will not continue.

You can learn more about this automated trading behavior [in our post ↗](#).

What is in the Important Links section of the documentation?

- [Governance Portal ↗](#): The site for Goldfinch's Governance proposals and Discussions. You can learn more about the Governance process in the [Governance section](#) of the documentation.
- [Developer Documentation ↗](#): A documentation site for developers and those auditing Goldfinch's code, for a deeper technical dive into Goldfinch's code, smart contract configuration, security, and other technical features.
- [Protocol Data Dash ↗](#)board: A Dune dashboard providing live insights into the Goldfinch protocol, including 30 day protocol revenue, repayment metrics, defaults, and more.
- [November 2021 Audit ↗](#): A link to the second protocol external protocol audit, completed by Trail of Bits.
- [Immunefi Bug Bounty ↗](#): The bounty for developer bug detection, focused on preventing code impacts such as loss of user funds and hosted by Immunefi.
- [Github ↗](#): The open-source repository for all of Goldfinch's code.

Important links

Connect